

Graph-Structured Models, Their Algorithms, and the Properties That Referee Them

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January 1, 2025

Abstract

A self-contained account of the problem classes this project supports—four primary ones and the instances derived from them—the algorithms that solve them, and—the part a methods section usually omits—the mathematical properties each algorithm is validated against, and why those properties are the right referees. Written for a reader with a scientific and performance-computing background but no background in phylogenetics or statistical physics. It describes *algorithms*, not an implementation: no result here depends on how any of it is coded, and the companion paper carries the results themselves.

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1 Notation

One symbol, one meaning, across this document and the companion paper.

Symbol	Meaning
$\mathcal{T}$	a tree topology
$\mathcal{G}$	a general undirected graph
$\mathcal{A}$	the state alphabet, $ \mathcal{A} = k$
ρ	the root of a rooted tree
$\mathcal{N}(\cdot)$	the neighbourhood a discrete move set reaches in one step
$\mathbf{Q}$	a rate matrix; $\mathbf{P}(t) = \exp(\mathbf{Q}t)$ its transition matrix
$\mathbf{S}$	a symmetric exchangeability matrix
π	a stationary or initial distribution
τ	branch lengths
$\boldsymbol{\theta}$	unconstrained parameters, $\boldsymbol{\theta} \in \mathbb{R}^d$
ϕ	the constraint map, $\phi(\boldsymbol{\theta})$ the feasible parameters
$\mathbf{D}$	the observed data
$\mathcal{L}$	the scalar being minimized
H	a Hamiltonian
$\mathbf{Z}$	a hidden state path
$\mathcal{E}$	an emission family
$\mathbf{H}$	a parity-check matrix; the codewords are the c with $\mathbf{H}c = 0$
H	the Sylvester–Hadamard matrix, italic where $\mathbf{H}$ is bold
x_i	a discrete variable of a factor graph, $x_i \in \mathcal{A}$
ψ_a	a factor: a non-negative function of the variables $x_{\partial a}$ it touches
$m_{a \rightarrow i}, m_{i \rightarrow a}$	a message from factor to variable, and from variable to factor
b_i, b_a	a belief: the marginal of a variable or a factor, exact on a tree
Z	a partition function; $\log Z$ the evidence
$T, \beta = 1/T$	a temperature and its inverse; a schedule is $T(t)$ over a budget of t steps

Two conventions are worth stating because they are load-bearing later. Optimization is always *minimization*, so a likelihood-based objective is a negative log-likelihood. And $\mathcal{L}$ is a function of $\boldsymbol{\theta}$, the unconstrained vector, not of the parameters a person names—the two are connected by ϕ and Section 2.2 explains why the distinction is not cosmetic.

A third convention governs the symbols the problem sections keep. Every model below is an instance of the factor graph of Section 2.1, and its variables are the x_i of that section; but a spin is written σ_i and a hidden state z_t , because those are the names a reader arrives with. The identification is made once, where each model is stated as a factor graph, and not repeated: $x_i \equiv \sigma_i$ on a Potts graph, $x_t \equiv z_t$ on a chain, $x_u \equiv z_u$ at a node of a tree. Every section then carries the same five parts, in this order: the model; the sizes it is supported at, stated from the fixtures and the size tiers; the model as a factor graph, with a sketch of that structure; the algorithms as instances of Equation 2 or of the optimization each is, stated in the appendix and cited from the section; and the validation, one referee at a time with what it does and does not establish. What differs between the problems is then visible as content and not as presentation, and a section missing one of the five parts fails a guard rather than a review.

2 The Shape All Four Problems Share

Each problem class below couples a *combinatorial* search over a structure to a *continuous* optimization of parameters given that structure. The two are not interleaved steps of one optimizer: a structural move changes what the parameter vector *means* and how long it is, so it constructs a new objective rather than taking a step inside an existing one. Everything else in this document follows from that division.

2.1 Factor Graphs: The One Structure

A tree under a substitution model, a Potts model on a graph and a hidden Markov chain are one object: a set of discrete variables $x_1, \dots, x_N$ and a set of factors ψ_a , each a non-negative function of a subset $x_{\partial a}$, with

$$p(x) = \frac{1}{Z} \prod_a \psi_a(x_{\partial a}), \quad Z = \sum_x \prod_a \psi_a(x_{\partial a}). \quad (1)$$

The bipartite graph between variables and factors is the factor graph [51, 46]. In its *normal* (Forney) form every variable is an edge between exactly two factors, a variable shared by more factors passing through an equality factor $\delta(x, x', \dots)$ [29]; the form changes the drawing and not the distribution, which is checked by computing the same marginals on both. Observations enter as factors: a leaf's state is an indicator, an emission is a unary factor over the hidden state, so the graph is over latent variables only and a density and a probability are handled alike.

Sum-product sends, from factor a to variable i ,

$$m_{a \rightarrow i}(x_i) = \sum_{x_{\partial a \setminus i}} \psi_a(x_{\partial a}) \prod_{j \in \partial a \setminus i} m_{j \rightarrow a}(x_j), \quad m_{i \rightarrow a}(x_i) = \prod_{b \in \partial i \setminus a} m_{b \rightarrow i}(x_i), \quad (2)$$

and on a tree one leaf-to-root and one root-to-leaf pass are exact (Appendix A.3): on the tree's factor graph this is pruning, Equation 18; on a chain it is the forward recursion, Equation 35; on a pairwise graph it is Equation 26. On a loopy graph the fixed point is the Bethe approximation of Equation 27. Replacing the sum by a maximum gives max-product, whose max-marginals carry the MAP assignment—on a chain, Viterbi. One implementation of Equation 2 therefore serves every problem class, and each is pinned to the evaluator that predates it.

A coupled shape. The coupled spatio-sequential model puts a Potts prior on spatial labels $\ell_n \in \{1..M\}$, one hidden chain $k_{s,m}$ per class, and a *gated* emission joining one label to one chain state:

$$\log p(\ell, k, x) = \beta \sum_{\langle n, n' \rangle} J_{nn'} \delta(\ell_n, \ell_{n'}) + \sum_m \left[\log \pi_m(k_{1,m}) + \sum_{s>1} \log A_m(k_{s-1,m}, k_{s,m}) \right] + \sum_n \sum_s \log P(x_{sn} \mid k_{s,\ell_n}, \theta_{\ell_n}) \quad (3)$$

which is Equation 1 with pairwise, chain and degree-two gated factors, and no fourth code path. Section 6 states the model in full, with the estimator its structure dictates.

A parity-check code. A low-density parity-check code is Equation 1 with binary variables, a unary factor per bit carrying the channel's log-likelihood ratio, and a hard parity constraint per row of the check matrix (Section 12). It is the problem message passing was invented for [31], and the fourth problem class: its decoder is Equation 2 specialised to one number per message, and is held to the general implementation on codes small enough for both.

What pins the general algorithm. Equation 2 is validated where each of its instances already has an oracle, and by nothing new: on a tree-shaped Potts graph against enumeration over $q^{|V|}$ configurations, on a chain against the sum over k^T paths, on a tree at one site against pruning, and on a loopy graph against the pairwise belief propagation of Equation 26—the same approximation reached by two codes, which is agreement and not truth. The Forney form is pinned by computing the same marginals on both drawings. A general algorithm that agreed only with itself would be a second opinion, not an evaluator (Section 13).

2.2 The Continuous Optimization Abstraction

Let the objective be a differentiable scalar $\mathcal{L}(\boldsymbol{\theta})$ over an unconstrained vector $\boldsymbol{\theta} \in \mathbb{R}^d$, with a map ϕ carrying $\boldsymbol{\theta}$ to the constrained parameters a model is stated in: positive branch lengths through a logarithm, a distribution over k states through a softmax on $k - 1$ free values, a positive scale through a logarithm.

Why the map and not a projection. An optimizer that must be *stopped* from leaving the feasible set will eventually leave it, and a constrained parameter that reaches its boundary has no interval around it worth reporting. Constraining by construction removes both failures: every point in $\mathbb{R}^d$ is feasible, and the gradient $\nabla_{\boldsymbol{\theta}} \mathcal{L}$ is defined everywhere.

Why the map must be injective. Where a reparameterization leaves $\mathcal{L}$ unchanged, the direction along it is not estimable: the observed information is singular and an interval is undefined. A softmax on k free values has exactly this flat direction—adding a constant to every value changes nothing—which is why $k - 1$ are used. Removing the freedom is the constraint map’s job, not the optimizer’s.

What follows for inference. Standard errors come from the observed information at the optimum, pushed through ϕ by the delta method: $\text{Var}(\phi(\boldsymbol{\theta})) \approx J \Sigma J^\top$ with J the Jacobian of ϕ . This is an *asymptotic* statement, exact only in the large-sample limit, which is why Section 13 makes realized interval coverage a measurement rather than an assumption.

2.3 Sampling the Posterior, and Adapting the Sampler

Where the delta-method interval is an approximation, the posterior can be sampled instead and an interval read as a quantile. Hamiltonian Monte Carlo [63] reads $\mathcal{L}$ as a negative log density, adds a momentum $p \sim \mathcal{N}(0, M)$, and proposes by integrating the Hamiltonian $H(\boldsymbol{\theta}, p) = \mathcal{L}(\boldsymbol{\theta}) + \frac{1}{2} p^\top M^{-1} p$ with a symplectic integrator for L steps of size ε , accepting with probability $\alpha = \min\{1, \exp(-\Delta H)\}$. The integrator is reversible and volume-preserving, so ΔH is the whole of the acceptance ratio; it is exact on the flow it conserves, so ΔH is small and bounded rather than growing with L . A temperature enters as the momentum’s variance, and a diagonal mass $M = \text{diag}(1/\sigma_i^2)$ enters as a change of coordinates, $\phi_i = \boldsymbol{\theta}_i/\sigma_i$: the unit-mass dynamics on $\mathcal{L}(\sigma \odot \phi)$ are the mass- M dynamics on $\boldsymbol{\theta}$ term for term, so one integrator serves every metric and every temperature.

Adaptation, as a warm-up. The step ε that is right for one posterior is wrong for the next, and the coordinates’ scales σ_i are the posterior’s. Both are set in a warm-up whose draws are discarded, after which the chain runs at fixed values and is a Markov chain with the target as its stationary distribution; a chain that kept adapting would not be. The scales are the sample standard deviations of a warm-up window. The step is driven to a target acceptance δ by dual averaging [41, Sec. 3.2]: after proposal m with acceptance probability α_m ,

$$\bar{h}_m = \left(1 - \frac{1}{m + t_0}\right) \bar{h}_{m-1} + \frac{\delta - \alpha_m}{m + t_0}, \quad \log \varepsilon_m = \mu - \frac{\sqrt{m}}{\gamma} \bar{h}_m, \quad \log \bar{\varepsilon}_m = m^{-\kappa} \log \varepsilon_m + (1 - m^{-\kappa}) \log \bar{\varepsilon}_{m-1}, \quad (4)$$

with $\mu = \log(10 \varepsilon_0)$ and the constants γ, t_0, κ setting the rate; the iterates ε_m do not converge, the averaged $\bar{\varepsilon}_m$ does, and the chain runs at the last of those. The same iteration is run twice, once at unit mass to estimate the scales and once on the metric they define, because a metric that has rescaled the coordinates has changed the step that is right for them.

Regime. On a target that is locally quadratic the leapfrog energy error is bounded and oscillatory, so $\alpha(\varepsilon)$ stays near one up to the stability limit $\varepsilon \omega_{\max} < 2$ on the stiffest direction and falls to zero past it: a target of $\delta = 0.65$ lies *on* that cliff, and an iteration driven by a single Metropolis probability per proposal, whose standard deviation is a sizeable fraction of one, oscillates across it and averages to a step whose acceptance is not δ . Two remedies, both measured rather than assumed: each proposal’s step is drawn uniformly from $\bar{\varepsilon}(1 \pm j)$ [63, Sec. 5.4.2.2], which averages α over the band and turns the cliff into a slope with one root; and γ is set larger than the value published for a trajectory-averaged statistic, since the iterates’ spread in $\log \varepsilon$ scales as $1/(\gamma\sqrt{m})$. The chain after warm-up keeps the jitter, since that is the step its acceptance was adapted for, and every draw is still from a kernel with the right stationary distribution.

A ladder from what it measures. The same principle sets a tempering ladder for Section 4.8. What makes a ladder right is the exchange acceptance of Equation 33 between each neighbouring pair—near zero the replicas are independent chains, near one a rung is redundant—so a warm-up measures it on a candidate ladder and revises: a pair below a stated band is bisected at the geometric mean, a rung both of whose pairs are above the band is removed, and a pair above the band beside one inside it has their shared rung moved geometrically halfway toward the far end of the pair inside. Geometric, because the exchange ratio depends on the temperatures through $\beta_i - \beta_j$, so for an energy whose spread is set by the temperature the acceptance is a function of the ratio of neighbours. The endpoints are the caller’s and are never moved; the warm-up’s cost in sweeps is charged to any comparison against a ladder set by hand.

What pins it. Equation 4 is arithmetic and is stepped by hand against the implementation; the mass matrix is a change of coordinates and is held to a mass-matrix leapfrog written out, to round-off. What a draw costs is the effective sample size per gradient evaluation, with the size estimated by the initial positive sequence [33] and that estimator held to an autoregressive chain whose integrated autocorrelation time is a closed form. The statistics are then three: the adapted chain’s acceptance lands within a stated distance of δ over independent seeds; its posterior means and spreads agree with a fixed-parameter chain’s within Monte Carlo error, on the analytic Gaussian and on a four-taxon tree; and the adapted ladder’s exchanges lie inside the band on the frustrated lattice, with its ground-state hit rate at equal sweeps reported beside the hand ladder’s. The fixed-parameter samplers are unchanged and remain the oracles.

2.4 The Discrete Search as a Decision Process

The structural search is a Markov decision process [87]: the state is the current structure with its fitted parameters, an action is a move from the neighbourhood $\mathcal{N}(\cdot)$, and the reward is the improvement in the objective. An episode ends on a step budget or when no move improves the score—a property of the state, and the same stopping rule the greedy baseline obeys, which is what makes the two comparable. Because the reward is a difference, the undiscounted return telescopes,

$$G = \sum_{t=0}^{T-1} R_t = \sum_{t=0}^{T-1} [\mathcal{L}(s_t) - \mathcal{L}(s_{t+1})] = \mathcal{L}(s_0) - \mathcal{L}(s_T), \quad (5)$$

the improvement between the first and last state whatever path joined them. That is what licenses an undiscounted return—a discount would prefer improvement found early over the same improvement found late—and it is why a truncation landing between a sacrifice and its payoff teaches the opposite of the truth, so a truncated episode is recorded as one.

The policy is a softmax over the scores of the *available* actions, $\pi(a \mid s) \propto \exp f(s, a)$ for $a \in \mathcal{N}(s)$, so a neighbourhood of any size is admissible. A scorer with a single weight on the improvement a move buys is an inverse temperature, and greedy search is its zero-temperature

limit; what such an agent can learn is exactly how much it should accept a worsening move to reach a better optimum, which is the credit-assignment question this search makes sharp. That class cannot rank two moves differently from hill climbing, so the tree environment also scores a move by what it computes without a fit: the change in parsimony score, the fraction of sites compatible with the split the move breaks and with the split it makes, and the sizes of the two subtrees it exchanges, each standardized within the neighbourhood. A feature constant over a neighbourhood is cancelled by the softmax and is excluded rather than carried; the Robinson–Foulds distance from the current tree, the same for every NNI neighbour, is the standing example. Whether the wider class learns more is measured at a matched budget, never inferred from its width. The policy is trained by the score-function estimator with a baseline,

$$\nabla J = \mathbb{E} \left[\sum_t \nabla \log \pi(a_t | s_t) (G_t - b(s_t)) \right], \quad G_t = \sum_{u \geq t} R_u, \quad (6)$$

where a baseline b independent of the sample it is subtracted from leaves the estimator unbiased and is there to reduce its variance, not its bias: G_t is a sum of objective differences whose scale is the problem’s, so an uncentred gradient varies by orders of magnitude between instances. The comparison against the greedy baseline is at a matched budget of objective evaluations per decision, never in wall-clock time.

What pins it. Equation 5 is an identity and is asserted exactly on every recorded episode. Equation 6 is an expectation, so its estimator is pinned against the gradient of the exact expected return, which is computable by enumerating every episode on a small environment, and the baseline’s unbiasedness is asserted by the same enumeration: the expected gradient with and without b agree. An environment small enough to enumerate is therefore the oracle for the learning machinery, exactly as a tree small enough to enumerate is the oracle for the search.

2.5 Actor–Critic and Proximal Policy Optimization

The baseline in (6) may depend on the state. The natural choice is the state value $V^\pi(s) = \mathbb{E}[G_t | s_t = s]$, the improvement the search still expects from s , estimated by a *critic* fitted to sampled returns; the estimator is then weighted by the *advantage* $A_t = G_t - V(s_t)$, unbiased for any critic and of least variance in this family when the critic is exact (Appendix B). Where the return is exact by enumeration, so is V^π , and a critic is scored against it rather than against its own loss. The Monte Carlo return and the one-step bootstrap $R_t + V(s_{t+1})$ are the two ends of the *generalized advantage* [82], at $\gamma = 1$,

$$A_t^{(\lambda)} = \sum_{l \geq 0} \lambda^l \delta_{t+l}, \quad \delta_t = R_t + V(s_{t+1}) - V(s_t), \quad (7)$$

with $V(s_T) = 0$ at a terminal state and the critic’s value at a truncated one; $\lambda = 1$ recovers $G_t - V(s_t)$ and $\lambda = 0$ the temporal difference, trading the critic’s bias against the return’s variance.

The score-function estimator is valid only for the policy that collected the episodes. *Proximal policy optimization* [83] reuses a batch for several gradient steps by importance-weighting each decision with $\rho_t = \pi(a_t | s_t) / \pi_{\text{old}}(a_t | s_t)$ and clipping the ratio’s effect on the objective,

$$\mathcal{L}_{\text{PPO}} = \mathbb{E} \left[\sum_t \min(\rho_t A_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) A_t) \right], \quad (8)$$

so that an update cannot move the policy far from the one the advantages were estimated under. At the collecting policy every ratio is one and, without clipping, the gradient of (8) is the actor–critic’s; the regression suite pins that identity, and measures on the Potts chain what the

reuse buys: the fraction of the 81 starts from which the trained policy reaches the enumerated optimum, at a matched number of episodes. The same ratio makes the update valid when a *behaviour* policy collected the episodes: with π_{old} replaced by β , the ϵ -greedy wrapper of the policy being trained, on an ϵ schedule, the search can leave a local optimum the softmax would settle into, and the estimator is corrected by π/β .

2.6 Planning with a Learned Prior and Value

A policy answers “which move” from what it has seen; a planner looks ahead. From the current state, each of n simulations descends a tree of successors choosing, at every expanded node, the child that maximizes

$$Q(s, a) + c \pi(a | s) \frac{\sqrt{N(s)}}{1 + N(s, a)}, \quad (9)$$

the mean return backed up through that child plus an exploration bonus that is the prior the policy gives the move, decaying with the child’s visits [84]; the first unexpanded child is scored by the critic and the return backed up to the root. The root’s visit counts, normalized, are a policy that improves on the prior, and *expert iteration* [6] trains the policy toward that distribution by cross-entropy and the critic toward the achieved return, so the planner’s answers become the policy’s. Every expansion evaluates a successor, so the search is counted in the unit the rest of this document counts, and the comparison with greedy states the rate and the evaluations together (Appendix B).

2.7 Surrogates and Bounds

The evaluation a discrete search pays for—a fit of τ per topology, or $\log Z$ for a lattice—is what makes the search expensive, so each problem carries *surrogates*: cheaper functions of the structure that stand in for it. A surrogate makes a claim, a lower bound, an upper bound, or a point prediction, and the claim is part of the object. An analytic bound is proved (Appendix C) and *certified*: checked against the exact value on every structure an oracle can score, with no violation allowed. A learned predictor is *calibrated* instead, shifted by a quantile of its held-out residuals, and certified to the coverage it claims. The search ranks a neighbourhood by a surrogate and fits only the top candidates (Algorithm 16); the accepted move is always evaluated in full, so a surrogate changes what is fitted and never what is reported.

For a topology the maximized log-likelihood $\ell^*(\mathcal{T}) = \max_{\tau} \ell(\mathcal{T}, \tau)$ is bracketed by two one-pass quantities. Any feasible lengths give a lower bound, and lengths $\hat{\tau}$ solved by non-negative least squares from the pairwise Jukes–Cantor distances $\hat{d}_{ab}$ are feasible:

$$\ell(\mathcal{T}, \hat{\tau}) \leq \ell^*(\mathcal{T}), \quad \hat{\tau} = \arg \min_{\tau \geq 0} \sum_{a < b} \left(\sum_{e \in \text{path}(a, b)} \tau_e - \hat{d}_{ab} \right)^2. \quad (10)$$

Under Jukes–Cantor $\mathbf{P}(t) = a \mathbf{I} + (1 - a) \mathbf{J}/k$ with $a = e^{-kt/(k-1)}$, so a site likelihood is multilinear in the a_e and its maximum over the box $[0, 1]^{|E|}$ sits at a vertex, where every branch either forbids a change or forgets the parent’s state at cost $1/k$. With the tree rooted at the first leaf, the vertex maximum of site s is $\pi(x_{1,s}) k^{-F_s(\mathcal{T})}$ for the site’s Fitch score F_s , whence

$$\ell^*(\mathcal{T}) \leq \sum_s \log \pi(x_{1,s}) - F(\mathcal{T}) \log k. \quad (11)$$

Both apply to a subtree as they do to the tree.

For a lattice the partition function is bracketed by two variational quantities. The naive mean-field bound is Gibbs’ inequality at a product distribution $q = \prod_i q_i$, tightest at its fixed point:

$$\log Z \geq \sum_i \sum_{\sigma} q_i(\sigma) h_i(\sigma) + \sum_{\langle i, j \rangle} J_{ij} \sum_{\sigma} q_i(\sigma) q_j(\sigma) - \sum_i \sum_{\sigma} q_i(\sigma) \log q_i(\sigma). \quad (12)$$

Because $\log Z$ is convex in the parameters, Jensen’s inequality over a distribution ρ on spanning trees T , with parameters $\theta(T)$ supported on T and averaging to θ , gives an upper bound each term of which is exact by one leaf-to-root pass:

$$\log Z(\theta) \leq \sum_T \rho(T) \log Z(\theta(T)), \quad \sum_T \rho(T) \theta(T) = \theta, \quad (13)$$

the tree-reweighted bound of Wainwright et al. [92]. From any bracket $L \leq \log Z(\beta) \leq U$ at inverse temperature β , and $e^{-\beta E_{\min}} \leq Z(\beta) \leq k^N e^{-\beta E_{\min}}$,

$$-\frac{U}{\beta} \leq E_{\min} \leq \frac{N \log k - L}{\beta}. \quad (14)$$

3 Problem Statement: Phylogenetic Inference

3.1 The model

Character evolution along a branch is a continuous-time Markov chain with rate matrix $\mathbf{Q}$; the transition probabilities over a branch of length t are the matrix exponential $\mathbf{P}(t) = \exp(\mathbf{Q}t)$ [27]. A time-reversible model factors as $\mathbf{Q} = \mathbf{S} \text{diag}(\boldsymbol{\pi})$ with $\mathbf{S}$ symmetric, and then satisfies detailed balance, $\pi_i \mathbf{Q}_{ij} = \pi_j \mathbf{Q}_{ji}$.

The simplest such model is Jukes–Cantor on k states: every substitution equally likely and $\boldsymbol{\pi}$ uniform, with $\mathbf{Q}$ scaled so that one unit of branch length is one expected substitution per site,

$$-\sum_{i \in \mathcal{A}} \pi_i \mathbf{Q}_{ii} = 1, \quad (15)$$

which puts the transition probabilities in closed form [27, 23]:

$$\mathbf{P}_{ij}(t) = \frac{1}{k} + \left(\delta_{ij} - \frac{1}{k} \right) \exp\left(-\frac{k}{k-1} t \right). \quad (16)$$

Rows sum to one, $\mathbf{P}(0) = \mathbf{I}$, and every entry tends to $1/k$ as $t \rightarrow \infty$; the general time-reversible model replaces the single rate by $\mathbf{S}$ and $\boldsymbol{\pi}$ and keeps Equation 15.

Given a topology $\mathcal{T}$ over n taxa, branch lengths $\boldsymbol{\tau}$, and an alignment $\mathbf{D}$ of L sites, sites evolve independently and the likelihood marginalizes every assignment z of states to the internal nodes:

$$p(\mathbf{D} \mid \mathcal{T}, \boldsymbol{\tau}, \mathbf{Q}) = \prod_{s=1}^L \sum_z \pi_{z_\rho} \prod_{(u \rightarrow v)} \mathbf{P}_{z_u z_v}(\tau_v), \quad (17)$$

the product over the branches $(u \rightarrow v)$ of the rooted tree with the leaves fixed by the data— k^{n-2} terms per site for a binary tree. Summing it directly is the oracle every faster evaluation of it is checked against.

Supported sizes. The declared range is $n \in [10, 1000]$ taxa by $L \in [100, 11,000]$ sites, and the largest evaluation measured is 200 taxa by 11,000 sites. What is *refereed* is smaller, because the referee is an enumeration: the $(2n - 5)!!$ unrooted topologies number 15 at five taxa, 105 at six, 945 at seven and 10,395 at eight, so the per-pull-request tier carries the five- and six-taxon fixtures, the stress tier seven, and the release tier eight and the twenty- and fifty-taxon trees that only the distance start referees. Direct marginalization over the k^{n-2} internal assignments of Equation 17 referees the likelihood on the same fixtures at $k = 4$ and L from 200 to 200,000 sites. The Hadamard start is offered from three to twelve taxa, its spectrum having 2^{n-1} entries, and every enumeration here refuses more than 200,000 configurations rather than attempting them.

3.2 As a factor graph

One site is one instance of Equation 1. The variables are the states $x_u \equiv z_u \in \mathcal{A}$ at every node u of the rooted tree, leaves included. The factors are the root prior $\psi_\rho(x_\rho) = \pi_{x_\rho}$, one pairwise factor per branch, $\psi_v(x_u, x_v) = \mathbf{P}_{x_u x_v}(\tau_v)$ for u the parent of v , and one indicator per leaf, $\psi_\ell(x_\ell) = \mathbb{I}[x_\ell = \text{observed}]$, which folds the data in. The bipartite graph is a tree, so Equation 2 is exact, and Z is the site likelihood $p(\mathbf{D}_s \mid \mathcal{T}, \boldsymbol{\tau}, \mathbf{Q})$; sites being independent (Equation 17), the alignment’s likelihood is the product of L such Z . In the Forney form every internal node’s state is an edge through an equality factor of degree three—two branches and, at the root, the prior—which is the drawing Section 6 uses. Figure 1 is that structure on four taxa.

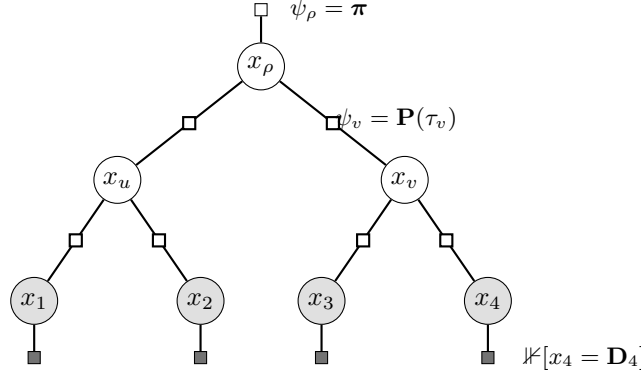


Figure 1: One site of a four-taxon phylogeny as a factor graph. Open circles are the states of the internal nodes and shaded circles the states at the leaves; small squares are factors. Each branch carries the pairwise factor $\mathbf{P}(\tau_v)$ of Equation 16, the root carries the prior π , and the filled square at each leaf is the indicator that folds in the observed character. The bipartite graph is a tree, so Equation 2 is exact and its leaf-to-root pass is Equation 18. A sketch of the structure, not a rendering of any instance: the drawn tree is four taxa, where the section’s supported sizes run from five to the taxon count enumeration reaches.

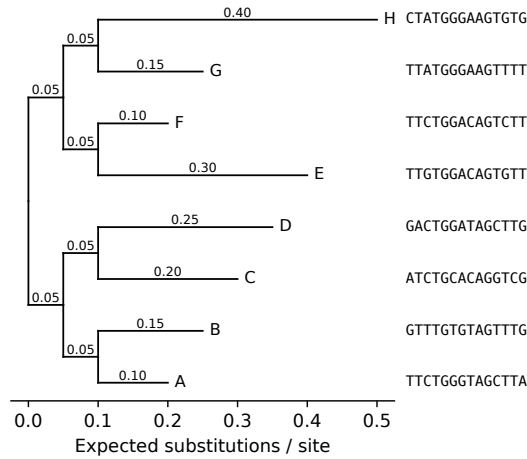


Figure 2: Simulated topology for 8 taxa under the Jukes-Cantor model (seed 20260904), branch lengths labelled in expected substitutions per site. The first 14 simulated sites are shown beside each leaf, in tree order, so the alignment the likelihood consumes can be read off against the topology that generated it.

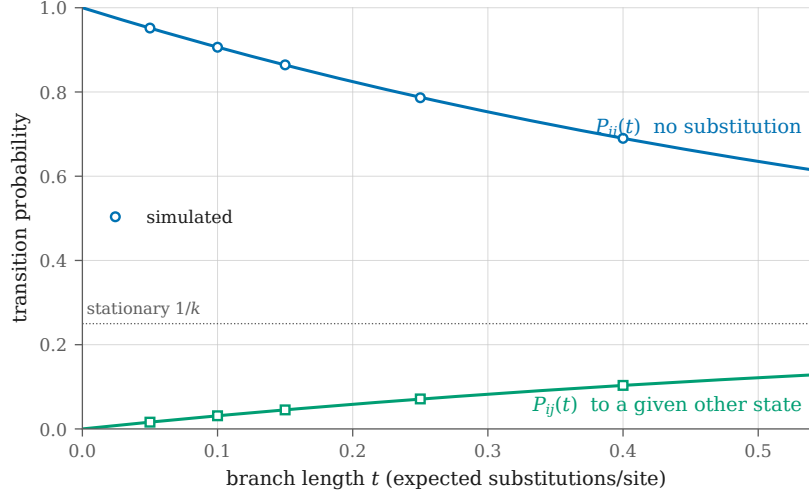


Figure 3: Closed-form Jukes-Cantor transition probabilities for $k = 4$ (curves) against the frequencies the simulator produced (markers), one pair per branch of the 5-branch fixture tree. Simulated at seed 20260902 over 200000 sites. Largest deviation $1.07\text{e-}03$, within the fixture’s declared Monte Carlo tolerance of $1\text{e-}02$. The simulator is validated against this analytic form, never against the likelihood code it feeds.

3.3 The algorithm: simulation

The generative direction of Equation 17 is a pre-order walk: draw the root state, then each node’s state from the row of its parent’s state. One site is Algorithm 1; L sites are L independent runs, and the empirical substitution frequencies along any branch converge to Equation 16, which is the check a simulator is held to—against the closed form, and never against the likelihood that will later be fitted to its output.

Algorithm 1 Forward simulation of one site under $(\mathcal{T}, \tau, \mathbf{Q})$

- 1: draw $z_\rho \sim \pi$
 - 2: **for** each non-root node v in pre-order, with parent u **do**
 - 3: draw $z_v \sim \mathbf{P}_{z_u, \cdot}(\tau_v)$
 - 4: **end for**
 - 5: **return** the leaf states $(z_\ell)_{\ell \in \text{leaves}(\mathcal{T})}$
-

3.4 The algorithm: pruning

Felsenstein’s pruning evaluates that marginal in $O(nLk^2)$ rather than $O(Lk^{n-2})$, by a post-order recursion carrying a partial likelihood $L_u(i)$ —the probability of the data below node u given that u is in state i :

$$L_u(i) = \prod_{v \in \text{children}(u)} \sum_{j \in \mathcal{A}} \mathbf{P}_{ij}(\tau_v) L_v(j), \quad (18)$$

with $L_u(i) = \mathbb{P}[u \text{ observed in state } i]$ at a leaf, and the site likelihood read off at the root,

$$p(\mathbf{D}_s \mid \mathcal{T}, \tau, \mathbf{Q}) = \sum_{i \in \mathcal{A}} \pi_i L_\rho(i) \quad (19)$$

[27, 23]; Appendix A.1 derives the recursion from the marginalization. Products of many small numbers underflow, so each node’s partials are rescaled and the logarithm of the scaling

accumulated separately; the rescaling must remain differentiable, since τ and $\mathbf{Q}$ are what the continuous optimization fits.

Equation 18 is Equation 2 run leaf to root with the root as the last variable: $L_u(i)$ is the product over u 's children of the factor-to-variable messages $m_{\psi_v \rightarrow u}(i)$, each of which sums the branch factor against the message arriving from below, and Equation 19 is the final factor-to-variable message from the prior. Replacing the sum by a maximum gives the most probable joint assignment of ancestral states; in the limit of short branches its cost counts substitutions, which is Fitch parsimony [27], the second criterion the search is scored against.

Both parsimony scores are the same post-order pass with no branch lengths. Fitch's recursion [28] carries at each node v the set S_v of states reachable at the minimum cost so far, and a count F of changes:

$$S_v = \begin{cases} S_a \cap S_b & S_a \cap S_b \neq \emptyset, \\ S_a \cup S_b & \text{and } F \leftarrow F + 1 \text{ otherwise,} \end{cases} \quad (20)$$

for the children a, b of v , folded pairwise at a trifurcation, a leaf's set being its observed state. Sankoff's recursion [78] replaces the set by a cost vector under a step matrix $W \in \mathbb{R}_{\geq 0}^{k \times k}$, W_{ij} the cost of a change from a parent in state i to a child in state j :

$$S_v(i) = \sum_{c \in \text{ch}(v)} \min_j [W_{ij} + S_c(j)], \quad S_{\text{leaf}}(i) = \begin{cases} 0 & i = x_{\text{leaf}}, \\ \infty & \text{otherwise,} \end{cases} \quad (21)$$

and a site's score is $\min_i S_{\text{root}}(i)$. It is Equation 18 in the min-plus semiring, and under the unit matrix $W = \mathbf{J} - \mathbf{I}$ it returns Fitch's count exactly. The score is that of a rooted tree: an asymmetric W scores each rooting of one unrooted topology differently, and a search over unrooted topologies is well defined only when W is a metric, where every rooting agrees.

Regime. Exact on every input, because a tree has no cycle. The one failure mode is numerical—underflow of a product of n factors each below one—and it is removed by the rescaling, not tolerated.

What pins it. Direct marginalization, Equation 17, at the sizes it allows, to a relative tolerance (Section 13.5). Two invariants past those sizes: the pulley principle, that a reversible model's likelihood does not depend on where the root is placed along a branch, and a gradient against central differences relative to its own norm. And the recovery test of Section 13: on simulated data the fitted branch lengths cover the truth at their nominal rate, and the search recovers the generating topology at a stated site count.

3.5 The algorithm: discrete search

The number of unrooted binary topologies on n taxa is $(2n - 5)!!$ [76], so exhaustive enumeration is an oracle at small n and nothing more. Two neighbourhoods are standard. *Nearest-neighbour interchange* (NNI) swaps the subtrees adjacent to one internal edge, giving $|\mathcal{N}_{\text{NNI}}(\mathcal{T})| = 2(n - 3)$. *Subtree prune and regraft* (SPR) detaches a subtree and reattaches it at another edge, giving $2(n - 3)(2n - 7)$ and containing NNI. Neither is complete in one step; both are complete in the transitive closure, which for NNI is the connectivity of the associated graph [27] and is inherited by SPR because it dominates NNI.

Hill climbing over either neighbourhood fits the continuous parameters of every candidate and moves to the best; its budget is counted in candidates scored, never in seconds, so a run reproduces from its seed. The same climb with Equation 20 or 21 in place of the fit is *large parsimony* [27]: nothing continuous is fitted, a candidate costs one pass, and the budget and the oracle are the same. A candidate that shares all but a few branches with its parent may start its fit from the parent's lengths, a branch being identified by the leaf bipartition below it

rather than by a node’s name; and a neighbourhood may be *ranked* by one evaluation at those lengths before the few best are fitted, the accepted move always in full, so every reported score is an optimum. Whether the ranking finds the fitted best is a property of the move set and is measured per move set, not assumed.

What pins it. Enumeration of the $(2n-5)!!$ topologies referees the search below eight taxa: the climb must reach the enumerated maximum, and the neighbourhood counts above are asserted exactly over every topology. A warm start is pinned by reaching the cold fit’s optimum; a cached partial likelihood by equalling the recomputed one bitwise, since it is the same arithmetic in the same order. Large parsimony is held to the same enumeration from every start, and on the Felsenstein-zone fixture its enumerated optimum is asserted to be the wrong tree while the likelihood optimum is the right one, because there the criterion’s failure is the theorem of Section 13 and not a defect of the search.

3.6 The initializer: distances, neighbor joining and the Hadamard spectrum

Spectral learning of a hidden Markov model [42] is a method of moments: the observable moments of short windows are inverted to the parameters, with a consistency guarantee and no likelihood surface to climb. Mossel and Roch [60] carry the same argument to a phylogeny—the same inversion of pairwise moments learns a nonsingular tree—and what it becomes in practice is a distance per pair of taxa and a tree from the distances. The pair co-occurrence matrix F of two taxa, F_{ij} the fraction of sites at which the first is in state i and the second in j , is the moment; under Equation 16 its off-diagonal mass is $\frac{k-1}{k}(1 - e^{-kt/(k-1)})$ for t the path length between the two, which inverts in closed form on the fraction p of differing sites,

$$\hat{d} = -\frac{k-1}{k} \log\left(1 - \frac{k}{k-1} p\right), \quad \text{Var}(\hat{d}) = \frac{p(1-p)}{L(1 - \frac{k}{k-1} p)^2}, \quad (22)$$

the variance by the delta method over the multinomial site counts. Under a general Markov process on the branches the log-det distance [86]

$$\hat{d} = -\frac{1}{k} \left[\log \det F - \frac{1}{2} (\log \det \Pi_x + \log \det \Pi_y) \right], \quad (23)$$

Π_x, Π_y the diagonal matrices of F ’s row and column sums, is additive along the tree whatever the process; it equals t when $\mathbf{Q}$ has trace $-k$, which Equation 15 gives the Jukes–Cantor model, and $-\text{tr}(\mathbf{Q})t/k$ under a general reversible $\mathbf{Q}$, a constant that leaves the tree unchanged.

Neighbor joining [77] is the tree from the distances, Algorithm 2: $n-3$ agglomerations of the pair minimizing $(n-2)d_{ij} - r_i - r_j$, r the row sums, each placing the new node by the three-point formula and reducing the matrix by one. On an additive matrix—one whose entries are the path lengths of some tree—it returns that tree with every branch length exact, and whether a matrix is additive is the four-point condition: of the three sums $d_{ij} + d_{kl}$, $d_{ik} + d_{jl}$, $d_{il} + d_{jk}$ over any four taxa, the two largest are equal [27]. On an estimated matrix Atteson [8] gives the guarantee: the topology is the true one whenever every entry’s error is below half the shortest branch, and Equation 22’s variance is what lets a run state whether it sits inside that radius.

For the symmetric two-state model the inversion is exact rather than pairwise. Hendy and Penny [40] index the 2^{n-1} site patterns by the set of taxa differing from a reference taxon, and the same subsets index the splits of the leaf set; with s the pattern frequencies and q the branch length on each split, zero where the tree has no such branch, the two are conjugate under the Sylvester–Hadamard matrix H of that order,

$$s = H^{-1} \exp(Hq), \quad q = H^{-1} \log(Hs), \quad (24)$$

Algorithm 2 Neighbor joining on a distance matrix D over n taxa

- 1: **while** more than three nodes remain **do**
 - 2: $r_i \leftarrow \sum_j D_{ij}$ for every remaining node i
 - 3: $(i, j) \leftarrow \arg \min_{i \neq j} (n - 2) D_{ij} - r_i - r_j$
 - 4: join i and j at a new node u with $D_{iu} = \frac{1}{2}D_{ij} + \frac{r_i - r_j}{2(n-2)}$, $D_{ju} = D_{ij} - D_{iu}$
 - 5: $D_{ku} \leftarrow \frac{1}{2}(D_{ik} + D_{jk} - D_{ij})$ for every other k ; remove i and j ; $n \leftarrow n - 1$
 - 6: **end while**
 - 7: join the last three at the root by the three-point formula
-

so the exact spectrum returns the tree’s own lengths on its splits and zero elsewhere. The *closest tree*, Algorithm 6, takes the $n - 3$ largest mutually compatible weights of an estimated q . A four-state Jukes–Cantor alignment reduces to the two-state model exactly by grouping the states in pairs, at $\frac{2}{3}$ of every branch length; the four-state conjugation proper is over the Klein four-group and is not implemented. Both estimators are a start and not an estimate: the fit and the search begin from them, and every reported number is the optimum they reach.

What pins it. Equation 22 against Equation 16 at the exact pair frequencies; Equation 23 against t under Jukes–Cantor and against $-\text{tr}(\mathbf{Q})t/k$ under a general reversible $\mathbf{Q}$; the stated variance by the coverage of its interval over seeds. Neighbor joining against the path lengths of every fixture and of random trees at 20 and 50 taxa, every branch to 10^{-12} ; on simulated alignments, recovery on every replicate inside Atteson’s radius, with no exception. The Hadamard conjugation against the pruning likelihood of Section 3 evaluated on every pattern, which shares nothing with the transform. And the starts against the objective’s own, at equal evaluations, through the budgeted comparison of Section 13.

3.7 Validation

Each referee below establishes one thing and is silent about the rest; the list is what the section may claim, and the paragraphs above say where each check sits.

- *Direct marginalization* over the k^{n-2} internal assignments, Equation 17, shares no traversal with Equation 18 and establishes that the recursion computes the marginal it claims, to a relative tolerance. It does not reach past the taxon count the supported sizes above give, and says nothing about the rescaling at lengths where the unscaled product would underflow.
- *The closed form* Equation 16 referees the simulator through the empirical substitution frequencies along a branch. It establishes that the generative direction is the model; it does not check the likelihood, and deliberately so, since a simulator checked against the evaluator fitted to its output referees nothing.
- *The pulley principle* — a reversible model’s likelihood is unchanged by where the root sits along a branch — and the gradient against central differences hold past every enumerable size. They are invariants, not answers: both are satisfied by a wrong likelihood that is wrong reversibly.
- *Topology enumeration* of the $(2n - 5)!!$ candidates referees the search, which must reach the enumerated maximum, and the neighbourhood counts $2(n - 3)$ and $2(n - 3)(2n - 7)$ are asserted exactly over every topology. It establishes that the climb found the best tree at those sizes; past them nothing here says the climb succeeded, which is what a branch-and-bound bound would supply.

- *Recovery of the simulated truth* — the generating topology at a stated site count, and branch lengths covered by their intervals at the nominal rate — is the referee where no oracle reaches. It establishes that the whole pipeline is calibrated on data from its own model; it establishes nothing about data the model does not describe.
- *The path lengths of a known tree* referee neighbor joining exactly on an additive matrix, and Atteson’s radius bounds when an estimated matrix still recovers the topology. Outside that radius the start is a start and not an estimate, and only the fit that follows it is reported.

4 Problem Statement: Potts Models in an External Field

4.1 The model

On a graph $\mathcal{G}$, each node i carries a spin $\sigma_i \in \{1, \dots, q\}$ and the energy is

$$H(\sigma) = - \sum_{\langle i, j \rangle} J_{ij} \delta(\sigma_i, \sigma_j) - \sum_i h_i(\sigma_i), \quad (25)$$

with couplings J_{ij} and an external field h_i [58]. The Boltzmann distribution is $p(\sigma) \propto \exp(-H(\sigma))$, and its normalizer sums over $q^{|V|}$ configurations.

Supported sizes. The exact referee is the $q^{|V|}$ enumeration, refused past 200,000 configurations, which puts the ceiling at eleven nodes at three states and seventeen at two. The fixtures sit under it: a chain of twelve sites at three states over 400 replicates, and an open 3×3 lattice at three states, whose $3^9 = 19,683$ configurations give the exact normalizer and every marginal. Above it the transfer matrix referees a strip of any length at a width that fits, belief propagation is compared on the 8×8 lattice where nothing is exact, the minimum cut is checked at extents 8, 12 and 16, and the cluster moves are separated from single-site sweeps at extents 8 and 24. The general graph carries $G(n, p)$ instances to a thousand nodes, refereed by the planted energy of Section 8 rather than by any enumeration.

4.2 As a factor graph

The variables are the spins, $x_i \equiv \sigma_i$. The factors are one unary $\psi_i(\sigma_i) = e^{h_i(\sigma_i)}$ per node and one pairwise $\psi_{ij}(\sigma_i, \sigma_j) = e^{J_{ij} \delta(\sigma_i, \sigma_j)}$ per edge, so Equation 1 is the Boltzmann distribution and Z its normalizer. At a temperature T the factors are raised to the power $\beta = 1/T$, which is the same graph with J and h scaled; nothing else about the model changes, and that is why a tempered chain can be held to the unscaled model’s enumeration. The graph is a tree exactly when $\mathcal{G}$ is, which is the regime every exact claim below is confined to. Figure 4 is that structure on the square lattice.

4.3 Why an odd cycle is the whole story

A two-state antiferromagnet ($J < 0$) wants every edge to disagree, which is possible exactly when $\mathcal{G}$ is 2-colourable. So every bipartite lattice—a chain, a square lattice—has a trivial, unfrustrated ground state and says nothing about a hard problem. A geometry containing odd cycles is what makes the ground state non-trivial, and where a counting argument closes exactly it fixes that ground state at every size. This is the difference between a fixture that exercises an optimizer and one that does not.

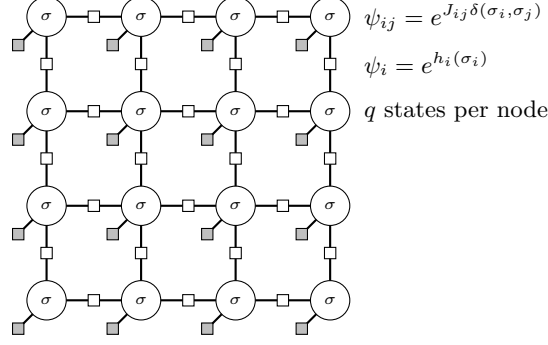


Figure 4: A Potts model in an external field as a factor graph, on the open 4×4 square lattice. Circles are the spins $\sigma_i \in \{1, \dots, q\}$; open squares on each edge are the pairwise factors of Equation 25 and shaded squares on a stub are the unary field factors. The graph has cycles, so Equation 2 on it is the Bethe approximation of Equation 27 and not the exact marginal; the square lattice is two-colourable, so it also carries no frustration, which is what Section 8 adds. A sketch of the structure and not a rendering of any instance.

4.4 The algorithm: belief propagation

Exact marginals come from enumeration at small sizes and from belief propagation on a tree, where it is exact; on a loopy graph it is an approximation whose accuracy is a property of the graph rather than of the implementation [46, 30]. It is Equation 2 with every factor pairwise or unary: composing the variable-to-factor message with the unary factor gives the node-to-node form, in which the message from i to a neighbour j is

$$m_{i \rightarrow j}(\sigma_j) \propto \sum_{\sigma_i} \exp(J_{ij} \delta(\sigma_i, \sigma_j) + h_i(\sigma_i)) \prod_{l \in \partial i \setminus j} m_{l \rightarrow i}(\sigma_i), \quad (26)$$

iterated to a fixed point, with beliefs $b_i(\sigma_i) \propto e^{h_i(\sigma_i)} \prod_{l \in \partial i} m_{l \rightarrow i}(\sigma_i)$ at a node and likewise on an edge. The messages are carried in the logarithm and normalized every sweep, because a coupling of a few units on a small lattice already puts a product of exponentials past what a linear recursion holds. The Bethe free energy of the fixed point,

$$F_{\text{Bethe}} = \sum_{\langle i, j \rangle} \sum_{\sigma_i, \sigma_j} b_{ij} (E_{ij} + \log b_{ij}) + \sum_i \sum_{\sigma_i} b_i E_i - \sum_i (d_i - 1) \sum_{\sigma_i} b_i \log b_i, \quad (27)$$

with $E_{ij} = -J_{ij} \delta(\sigma_i, \sigma_j)$, $E_i = -h_i(\sigma_i)$ and d_i the degree of i , equals $-\log Z$ exactly on a tree and approximates it on a loopy graph [97, 58]; Appendix A.4 states why, and Appendix A.3 places the tree case. A fixed point the iteration never reached is not an estimate of anything, so non-convergence is a refusal rather than a number.

Regime. Exact on a tree, in two passes. On a loopy graph the flooding schedule is iterated with damping to a tolerance and returns the Bethe approximation, and the tree schedule is refused rather than run, because a number that looks exact and is not is the worst output an evaluator can produce.

What pins it. On a tree, $\log Z$ and every marginal against enumeration to a stated relative tolerance. On a loopy graph nothing asserts agreement with the truth—that would assert something false—and what is asserted is the structure the physics requires: exact at zero coupling, and a deviation from the exact transfer matrix that grows away from it and is reported as a measurement.

4.5 The algorithm: ground states as cuts

With two states write $\sigma_i \in \{-1, +1\}$. The pairwise term of Equation 25 is constant on agreeing edges and pays J_{ij} on disagreeing ones, so minimizing H minimizes the weight of the *cut*—the set of edges whose endpoints differ—plus the field:

$$H(\sigma) = \sum_{\langle i,j \rangle: \sigma_i \neq \sigma_j} J_{ij} - \sum_i h_i(\sigma_i) + \text{const.} \quad (28)$$

When every $J_{ij} \geq 0$ the pairwise term is submodular, the field becomes a terminal edge to a source or a sink, and the ground state is a minimum s – t cut, exact by max-flow in polynomial time [37, 21]. When couplings are negative the problem is maximum cut, which is NP-hard; the semidefinite relaxation rounded by a random hyperplane gives a cut whose expected weight is at least $\alpha_{\text{GW}} = 0.87856$ of the optimum,

$$\mathbb{E}[w(\text{cut})] \geq \alpha_{\text{GW}} \max_{\sigma} w(\sigma), \quad \alpha_{\text{GW}} = \min_{0 < \vartheta \leq \pi} \frac{2}{\pi} \frac{\vartheta}{1 - \cos \vartheta}, \quad (29)$$

and the relaxation’s own value bounds the optimum from above [34]. The bound is the claim that holds at every size; the realized ratio on an instance is measured beside it and is never quoted as the guarantee.

Regime. Exact where the couplings are submodular and refused rather than approximated where they are not (Section 13): a wrong ground state on a structured instance is indistinguishable from a right one at any size worth solving. The relaxation runs on any instance and certifies a ratio, not an optimum.

What pins it. Enumeration over $2^{|V|}$ configurations where it reaches; where it does not, a *reduction*: on a bipartite graph the maximum cut is every edge, on a uniform ferromagnet the ground state is constant, and a construction error in the cut graph yields a labelling that is merely worse, which only a regime where the general construction must reproduce a simpler validated one will catch. A fixture whose answer is trivial measures nothing, so the odd cycles of Section 4 are what the instances carry.

4.6 The algorithm: alpha expansion

With $q > 2$ states the ground state is NP-hard in general, and alpha expansion approximates it by a sequence of exact two-state problems [18]. An *expansion* on label α lets every site keep its label or switch to α ; the best such move is the minimum cut of a two-state graph built from the current labelling, exact when the pairwise term is a metric on labels, which $\delta(\sigma_i, \sigma_j)$ is. Cycling over α until no expansion lowers the energy, Algorithm 12, reaches a local minimum within a constant of the global one,

$$H(\hat{\sigma}) \leq 2c H(\sigma^*), \quad c = \max_{\alpha \neq \beta'} d(\alpha, \beta') / \min_{\alpha \neq \beta'} d(\alpha, \beta'), \quad (30)$$

so a factor of two for the Potts metric, where $c = 1$. The guarantee assumes each expansion is solved to optimality; where it is solved approximately the certificate is optimistic, and what can be asserted is the symptom—a value an exact solve could not have produced.

What pins it. The two-label case must reproduce the exact minimum cut of Section 4.5 exactly; enumeration referees the multi-label case where it reaches; and the realized ratio against the enumerated optimum is measured, since the bound is a theorem and the ratio is a number.

4.7 The algorithm: sampling

The single-site heat bath draws each spin from its conditional,

$$p(\sigma_i \mid \sigma_{\partial i}) \propto \exp\left(\beta \left[\sum_{j \in \partial i} J_{ij} \delta(\sigma_i, \sigma_j) + h_i(\sigma_i) \right]\right), \quad (31)$$

one site at a time; a *sweep* is one update of every site, Algorithm 8. Near a critical coupling correlated regions span the lattice and single-site moves decorrelate slowly. Cluster moves flip such regions together [90, 95, 64]: in the Edwards–Sokal representation [25] a bond is placed between equal neighbours with probability $1 - e^{-\beta J_{ij}}$, and each bond-connected cluster is recoloured uniformly—every cluster at once in Swendsen–Wang, one cluster grown from a random seed in Wolff. The bond construction is exact at zero field only. In a field the recolouring is accepted with the Metropolis probability on the field energy alone,

$$a = \min \left\{ 1, \exp\left(\beta \sum_{i \in \mathcal{C}} [h_i(\nu) - h_i(\mu)]\right) \right\}, \quad (32)$$

for a cluster $\mathcal{C}$ recoloured from μ to ν , which restores detailed balance (Appendix A.5). Algorithm 9 is the construction and Algorithm 10 the replica exchange of Section 4.8 over it.

Regime. Every move preserves the Boltzmann distribution at β , so a sweep may not stop on a state-dependent condition: composing a *number* of steps chosen from the outcome biases toward the states that triggered the stop. The bond probability is a probability only for $J_{ij} \geq 0$, and an antiferromagnet has no like-spin clusters to flip, so the cluster moves refuse a negative coupling rather than run badly.

What pins it. A sampler is validated by the distribution it converges to and never by inspection: at an enumerable size the exact Boltzmann distribution is available, and a chain is held to it by a goodness-of-fit test at a declared significance, thinned by several autocorrelation times because successive sweeps are not independent draws. The test’s power is itself pinned—a move set with the field accept step removed runs, mixes, and is rejected.

4.8 Temperature, annealing and tempering

A temperature T scales the model, H/T , so a chain at T targets $\exp(-H/T)$; a *schedule* $T(t)$ over a budget of t steps is one object, shared by every consumer of it, with both endpoints reached exactly at the declared steps and a step past the end refused. Simulated annealing is the sampler on a decreasing schedule, returning the lowest-energy state seen [45]; at $T \rightarrow 0$ the heat bath is the argmin over each site’s conditional, which is iterated conditional modes, so annealing and greedy descent are one search separated by the schedule alone.

Parallel tempering runs R replicas at fixed temperatures $T_1 < \dots < T_R$ and, after each sweep, proposes to exchange the configurations of adjacent replicas, accepting with

$$a_{ij} = \min \left\{ 1, \exp[(\beta_i - \beta_j)(E_i - E_j)] \right\}, \quad (33)$$

the ratio of the joint target, a product of tempered marginals [89, 32, 24]; Appendix A.6 derives it. The hot replicas cross barriers the cold one cannot, and an exchange carries what they find down the ladder. Each replica draws from its own generator, spawned from one parent that then draws only the exchange uniforms, so no two replicas share a stream and one seed reproduces the run.

The replica at $T_1 = 1$ targets the unscaled model, so the fraction of its N recorded sweeps spent at a structure x estimates that structure’s weight over the whole space,

$$\hat{w}(x) = \frac{1}{N} \sum_{n=1}^N [\sigma_1^{(n)} = x] \rightarrow \frac{\exp(-H(x))}{Z}, \quad (34)$$

where a weight over a neighbourhood is conditional on that neighbourhood and enumeration is refused past its cap. The structure may be a labelling, a hidden path, or a topology with H the negative fitted log-likelihood, in which case the limit is the flat-prior weight over maximized likelihoods and not a marginal over branch lengths. It is a Monte Carlo estimate, and is named a posterior weight only beside the exchange acceptance of every adjacent pair and the autocorrelation time of the indicator, which say whether the ensemble mixed.

What pins it. Every replica’s marginal must still be $\exp(-H/T_r)$, enumerated from the *unscaled* model, with exchanges on: a wrong exchange ratio contaminates the cold replica with hot configurations and only that test says so. Its power is pinned by the paired negative—an exchange that omits the energy term still runs and still mixes, and is rejected on every replica. Whether tempering or annealing beats restarts of descent at an equal budget of sweeps is not a property of the algorithms but of the instance, and is measured on the frustrated, past-enumeration instances the roadmap needs, never asserted in general.

4.9 Validation

- *Enumeration of the $q^{|V|}$ configurations* referees $\log Z$, every marginal, and the ground state at the sizes above. It establishes exactness there and nothing at all past them, which is why every claim about a larger lattice is either a structural invariant or a measurement.
- *The transfer matrix* referees the normalizer of a strip independently of the message passing, so a chain and a narrow lattice have an oracle past enumeration. It establishes agreement at a width, not a bound on the error at any width.
- *Zero coupling* is where belief propagation is exact on any graph. It establishes that the loopy implementation reduces correctly; the deviation away from it is reported as a measurement and asserted against nothing, because asserting agreement with the truth on a loopy graph would assert something false.
- *A minimum cut* referees the two-state submodular ground state exactly at any size max-flow reaches, and alpha expansion must reproduce it at two labels. It establishes the multi-label solver’s construction; it does not establish the multi-label optimum, whose bound Equation 30 states as a theorem while the realized ratio is measured beside it.
- *A goodness-of-fit test against the enumerated Boltzmann distribution*, thinned by several autocorrelation times, referees a sampler. It establishes the distribution the chain converges to at an enumerable size; it establishes no mixing time, and its power is itself pinned by a paired negative — the same move set with the field accept step removed runs, mixes, and is rejected.
- *The unscaled model’s enumeration*, with exchanges on, referees every replica’s marginal in parallel tempering. It establishes that the exchange ratio does not contaminate the cold replica; whether tempering beats restarts at an equal budget is a property of the instance and is measured on the frustrated instances of Section 8, never asserted in general.

5 Problem Statement: Hidden Markov Models

5.1 The model

A hidden path $\mathbf{Z} = (z_1, \dots, z_T)$ over k states evolves by an initial distribution $\boldsymbol{\pi}$ and a transition matrix $\mathbf{A}$; each position emits an observation from an emission family $\mathcal{E}$ indexed by its state. The evidence marginalizes k^T paths.

Supported sizes. The exact referee is that k^T path enumeration, refused past 200,000 paths, so the oracle reaches eleven positions at three states and seventeen at two. The fixture is three states, four symbols and fifteen positions over 600 sequences: long enough that a fit recovers the transition matrix, short enough that any window of it enumerates. Six emission families run against the same recursions, and the recursions themselves are exact at every length, the chain being a tree; the release tier fits sequences past the enumerable length and is refereed there by recovery of the simulated parameters alone, up to the permutation of hidden states.

5.2 As a factor graph

The variables are the hidden states, $x_t \equiv z_t$. The factors are $\psi_1(z_1) = \pi_{z_1} \mathcal{E}_{z_1}(y_1)$ at the first position and, for $t > 1$, one pairwise $\psi_t(z_{t-1}, z_t) = \mathbf{A}_{z_{t-1}z_t}$ per transition and one unary $\mathcal{E}_{z_t}(y_t)$ per emission: the observation y_t is folded into the table, so a probability and a density enter the same way and the graph is over latent variables only. The chain is a tree, and Z is the evidence $p(\mathbf{D})$. An interior state sits on three factors, so in the Forney form it passes through an equality node—the drawing Section 6 uses for each class’s chain. Figure 5 is that structure on five positions.

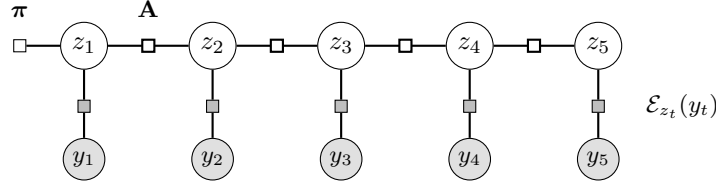


Figure 5: A hidden Markov model as a factor graph. Open circles are the hidden states z_t over k values, open squares the transition factors $\mathbf{A}$, the leftmost square the initial distribution π , and shaded squares the emission factors of the family $\mathcal{E}$, each folding in one observation y_t (shaded circle). The chain is a tree, so Equation 2 is exact and its two passes are Equation 35 and Equation 36. Removing the transition factors leaves the Gaussian mixture of Section 9; keeping them and gating the emissions on a spatial label gives the coupled model of Section 6. A sketch of the structure and not a rendering of any instance.

The emission family is the only part that knows what an observation is. Everything else—the recursions, the expectation step, path enumeration—needs three things from it: draw an observation given a state, score one, and re-estimate itself from posterior weights. Separating it is what lets one set of recursions serve a finite alphabet, a real observation, and a count.

A density is not a probability. For a family over a countable alphabet the evidence is a probability, so $\log p(\mathbf{D}) \leq 0$. For a family over the reals it is a *density* and carries no such bound. A check written against the first is false under the second, so what may be asserted is keyed on the support the model declares.

5.3 The algorithm: forward–backward

The forward recursion is Equation 18 on a chain:

$$\alpha_t(i) = \left[\sum_j \alpha_{t-1}(j) \mathbf{A}_{ji} \right] \mathcal{E}_i(y_t), \quad p(\mathbf{D}) = \sum_i \alpha_T(i). \quad (35)$$

Pruning on a caterpillar tree carrying one observed leaf per internal node *is* this recursion [23, 46]: the three problem classes are one marginalization over three structures, not three

unrelated models sharing an optimizer; Appendix A.2 derives both passes from it. In the terms of Equation 2, α_t is the message arriving at z_t from the left multiplied by the emission factor, the backward recursion β_t is the same pass from the right, and the posterior at a position is their normalized product,

$$p(z_t = i \mid \mathbf{D}) = \frac{\alpha_t(i) \beta_t(i)}{p(\mathbf{D})}, \quad \beta_t(i) = \sum_j \mathbf{A}_{ij} \mathcal{E}_j(y_{t+1}) \beta_{t+1}(j), \quad (36)$$

which is the belief b_t of Section 1. Max-product on the same chain is Viterbi,

$$\delta_t(i) = \left[\max_j \delta_{t-1}(j) \mathbf{A}_{ji} \right] \mathcal{E}_i(y_t), \quad (37)$$

with the path read back from the argmax at each step; on a chain with a unique maximum the max-marginals' argmax at every position is that path [23].

Regime. Exact for every length, the chain being a tree. The one hazard is the same underflow pruning meets, and the same remedy applies: the recursion runs in the logarithm.

What pins it. The sum over every one of the k^T paths, written out term by term with no recursion, referees the evidence, the posteriors and the Viterbi path at the lengths it allows; a gradient against central differences past them. The enumeration is the oracle two decoders are separated by, so it cannot be validated by a decoder, and it is pinned in turn against algebra: quantities that can be worked out by hand on a two-state, two-step chain.

5.4 The algorithm: expectation-maximization

Baum–Welch, Algorithm 14, alternates a posterior over hidden states given the parameters with a re-estimate of the parameters given that posterior, and increases the likelihood at every iteration [55]. The initial distribution and the transitions re-estimate in closed form for every family; the emission re-estimate is the family's own, and *whether it is a formula* is the property that separates the families. A weighted mean is; a dispersion parameter whose score involves the digamma function is not, and its M step is itself an optimization—which is the case that says whether an interface built around closed forms was built correctly.

What pins it. Monotonicity, asserted at every iterate; agreement of the fixed point with a gradient fit of the same likelihood; and recovery of simulated parameters up to the permutation of hidden states, with the alignment part of the test (Section 13). Where the family's M step is itself an optimization, its refusals at the boundary of the parameter space are pinned too: a dispersion the data cannot identify is refused, not clamped.

5.5 Two decodings, and they are different answers

Viterbi returns the single path of highest joint probability; posterior decoding returns, at each position independently, the state of highest marginal probability. Where they disagree, the posterior-decoded sequence can be a path the model assigns zero probability to, because nothing constrains consecutive choices to be compatible. They agree on almost every instance, which is exactly why an implementation that computes one and reports the other survives a test suite with no case where they diverge. Equation 37 and the argmax of Equation 36 are those two decodings; the fixture on which they differ is constructed and kept, because it is the only kind of case that can tell an implementation of one from an implementation of the other. Sampling the path posterior rather than decoding it is Algorithm 11, the block sweep the coupled model's E step also uses.

5.6 Validation

- *The sum over all k^T paths*, written out term by term with no recursion, referees the evidence, both decodings and every posterior at the lengths above. It establishes the recursions exactly; it is the oracle that separates the two decodings, so it cannot itself be checked by a decoder, and it is pinned in turn against quantities worked out by hand on a two-state, two-step chain.
- *Central differences* referee the gradient past the enumerable length. They establish that the differentiable path agrees with the recursion’s own value; they establish nothing about that value.
- *Monotonicity* referees expectation–maximization at every iterate. It establishes that the M step is an ascent, which is the one property the algorithm guarantees; it does not establish that the fixed point is the maximum, which is why the fixed point is also required to agree with an independent gradient fit of the same likelihood.
- *Recovery up to the permutation of hidden states* is the referee where no oracle reaches. It establishes calibration on data from the model, with the alignment an explicit part of the test rather than an accident of labelling; on a family whose M step is itself an optimization it also pins the refusals at the boundary, a dispersion the data cannot identify being refused and not clamped.
- *The declared support* decides what may be asserted at all: for a family over a countable alphabet the evidence is a probability and $\log p(\mathbf{D}) \leq 0$; for a family over the reals it is a density and carries no such bound. A check written against the first is false under the second.

6 Problem Statement: The Coupled Spatio-Sequential Model

6.1 The model

Observations x_{sn} are indexed by a spatial node n of a graph $\mathcal{G}$ and a sequential position $s = 1, \dots, S$. Each node carries a class label $\ell_n \in \{1, \dots, M\}$; each class m carries its own hidden chain $k_{s,m} \in \{1, \dots, K\}$; and the observation at (s, n) is emitted by the chain of the class the node belongs to, under that class’s emission parameters θ_m . The joint is Equation 3. Its three terms are the three problem classes above: a Potts prior on the labels,

$$\log p(\ell) = \beta \sum_{\langle n, n' \rangle} J_{nn'} \delta(\ell_n, \ell_{n'}) - \log Z_{\text{Potts}}, \quad J_{nn'} \geq 0, \quad (38)$$

ferromagnetic so that neighbouring nodes prefer one class—the penalty on disagreement, $-\beta \sum J_{nn'} \mathbb{I}[\ell_n \neq \ell_{n'}]$, is the same distribution up to a constant; one hidden Markov chain per class, in the simplest case with a circulant transition,

$$\mathbf{A}_m(i, j) = t \delta_{ij} + \frac{1-t}{K-1} (1 - \delta_{ij}), \quad (39)$$

whose self-transition rate t and initial distribution Π_m are fitted; and a *gated* emission, $\mathbb{I}[\ell_n = m] \log p(x_{sn} | k_{s,m}, \theta_m)$, which is what couples the two. Marginalizing k recovers a Potts model in a field the chains define; marginalizing ℓ recovers a mixture of hidden Markov models. Neither marginal is tractable on its own, and the structure of the joint is what dictates the estimator.

Supported sizes. The joint has $M^N K^{SM}$ states, so the enumerable instance is small: the canonical one is a 2×2 open lattice at $M = K = 2$ over $S = 6$ positions, $2^4 \cdot 2^{12} = 65,536$ joint states, inside the 200,000-state refusal and with its emissions separated far enough that a draw at $S = 6$ identifies the classes. Past it the model runs on the lattices of Section 4 at the sizes stated there, with the number of classes and chain states in single figures and the number of positions in the tens, and is refereed by the planted labels alone — which is the cell Section 11 marks as wanting an oracle.

6.2 As a factor graph

The variables are the N labels and the SM chain states. The factors are the pairwise Potts factors of Section 4 on the labels, the chain factors of Section 5 on each class’s states, and one gated emission factor per (n, s, m) over the pair $(\ell_n, k_{s,m})$, equal to $p(x_{sn} | k_{s,m}, \theta_m)$ when $\ell_n = m$ and to one otherwise. A label sits on every emission factor of its node and on every Potts factor of its edges; in the Forney form it passes through an equality node of that degree, and the gated factors are the higher-degree nodes that join the lattice to the chains (Figure 6). The graph has cycles, so Equation 2 is the Bethe approximation on it, and the exact quantities below come from enumeration on an instance small enough to enumerate—a 2×2 lattice with $M = 2$, $K = 2$, $S = 3$ has $2^4 \cdot 2^{12}$ joint states.

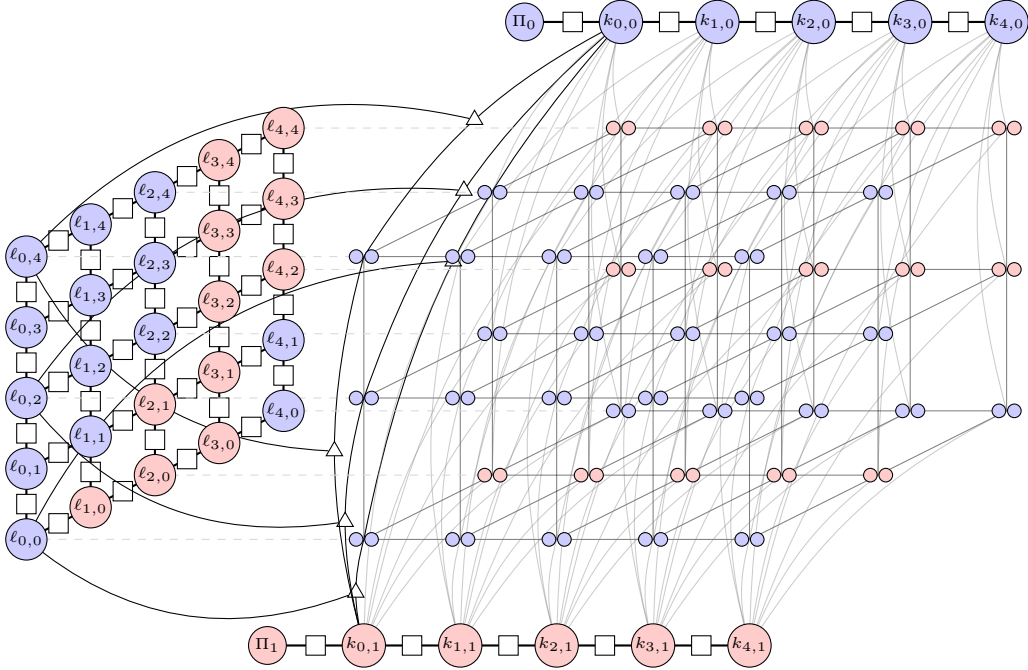


Figure 6: The coupled spatio-sequential model as a Forney-style factor graph. Each spatial node n carries a class label $\ell_n \in \{1, \dots, M\}$ (circles on the lattice, coloured by class for $M = 2$); each class m carries a hidden chain $k_{s,m} \in \{1, \dots, K\}$ with initial distribution Π_m (the two chains at the sides). Pairwise factors (squares) carry the Potts prior on the labels, Equation 38, and the chain prior, Equation 39. The gated emission factors (triangles) join one label to one chain state and contribute $\log p(x_{sn} | k_{s,m}, \theta_m)$ only when $\ell_n = m$; the observations x_{sn} are the small circles along each column. Every second observation column is drawn along each spatial axis, and an illustrative subset of the gated factors, so the structure is legible.

6.3 The estimator: block-coordinate ascent

The blocks are the emission parameters θ and the labels ℓ , with the chain states marginalized. For θ given ℓ , the gradient of the marginal likelihood is the posterior-weighted gradient of the emission terms,

$$\nabla_{\theta} \log p(x \mid \ell, \theta) = \sum_{n,m} \mathbb{K}[\ell_n = m] \sum_{s,k_{s,m}} \mathbb{Q}(k_{s,m} \mid \theta, \ell; x) \nabla_{\theta} \log p(x_{sn} \mid k_{s,m}, \theta_m), \quad (40)$$

where $\mathbb{Q}$ is the per-class posterior of Equation 36: this is expectation–maximization with the E step forward–backward on each class’s chain over the nodes assigned to it, and t and Π_m re-estimate in closed form. For ℓ given θ , the same posterior defines a per-node, per-class external field,

$$H_{nm}(\theta, \theta'; \ell) \equiv - \sum_{s,k_{s,m}} \mathbb{Q}(k_{s,m} \mid \theta', \ell; x) \log p(x_{sn} \mid k_{s,m}, \theta), \quad (41)$$

and the label update is the ground state of a Potts model in that field,

$$\ell^* = \arg \min_{\ell} \beta \left[\sum_{\langle n, n' \rangle} J_{nn'} \mathbb{K}[\ell_n \neq \ell_{n'}] + \sum_{n,m} \mathbb{K}[\ell_n = m] H_{nm} \right], \quad (42)$$

which is the problem of Section 4.6, solved by alpha expansion with its bound, descended by iterated conditional modes, or sampled by an annealed cluster move with a field (Section 4.7). Each block does not decrease the joint likelihood, which is the monotonicity every expectation–maximization here is held to.

Why a cluster move. Under an effective spatial prior each class spans a contiguous region, and a single-site move must cross a region one node at a time against the prior’s penalty; iterated conditional modes freezes into spurious disjoint clusters, and single-site annealing slows critically for the same reason. The Wolff move of Section 4.7 recolours a whole cluster at once, and Algorithm 3 is that move with the field of Equation 41 in its accept step. At $\beta \rightarrow 0$ it is a single-site Gibbs update; at $\beta \gg 1$ it is the greedy entrapment of iterated conditional modes; annealing in β moves between them.

Algorithm 3 One Wolff update on the labels, in the external field

- 1: draw a root ρ uniformly from the nodes; $\mu \leftarrow \ell_{\rho}$; $\mathcal{C} \leftarrow \{\rho\}$; queue $Q \leftarrow \{\rho\}$
 - 2: **while** $Q \neq \emptyset$ **do**
 - 3: pop n from Q
 - 4: **for** each neighbour n' of n with $\ell_{n'} = \mu$ and $n' \notin \mathcal{C}$ **do**
 - 5: with probability $1 - e^{-\beta J_{nn'}}$: add n' to $\mathcal{C}$ and to Q
 - 6: **end for**
 - 7: **end while**
 - 8: draw a new label ν uniformly from $\{1, \dots, M\}$
 - 9: $\Delta H \leftarrow \sum_{n \in \mathcal{C}} (H_{n\nu} - H_{n\mu})$
 - 10: with probability $\min\{1, e^{-\beta \Delta H}\}$: $\ell_n \leftarrow \nu$ for every $n \in \mathcal{C}$
-

6.4 The initializer: data-sampled burn-in

A block ascent on this joint is only as good as where it starts, and a start that ignores the graph and the emission family—*k-means++* on the raw observations, say—leaves degenerate classes and false positives that the ascent then polishes. Algorithm 4 anneals in β from zero, so that early on the labels are nearly free and the chains and emission parameters are fitted to what the data alone supports, and the spatial prior is tightened only as the classes separate. Its seeding

step, EMISSIONMIXTURE++, is k -means++ with the emission family’s own negative log-density in place of squared Euclidean distance [7]: candidates are drawn proportionally to how badly the current parameters explain them, constrained to the observations the current labels and states assign to the class being seeded. The $8(\ln K + 2)$ expected-cost bound of k -means++ is a statement about squared distance and is not inherited; what is measured instead is recovery against a uniform start at an equal budget, over seeds, reported whether it wins or not.

Algorithm 4 Data-sampled burn-in on the coupled graph

```

1:  $\beta \leftarrow 0$ ;  $H \leftarrow 0$ ;  $\ell_n \sim \text{Uniform}\{1, \dots, M\}$  for every node
2:  $\theta \leftarrow \text{EMISSIONMIXTURE++}(\beta, K, \theta, x)$ 
3: while  $\beta < \beta_0$  do
4:   WOLFFUPDATE( $\beta, \ell, J, H$ ) ▷ Algorithm 3
5:   for each class  $m$  do draw the chain  $k_{\cdot, m}$  by block Gibbs given  $\ell, \theta$ 
6:   end for
7:   for each state  $k$  do  $\theta_k \leftarrow \text{EMISSIONMIXTURE++}(\beta, K, \theta, x_k, \ell)$  ▷ state-constrained
8:   end for
9:    $\mathbb{Q} \leftarrow$  forward-backward per class (Equation 36); M step for  $\theta, t, \Pi$ 
10:   $H_{nm} \leftarrow$  Equation 41
11:   $\beta \leftarrow \beta + \Delta\beta$ 
12: end while
13: return  $\ell, \theta$ 

```

What pins it. The graph of Equation 3 is constructed from the three adapters and its log-density equals the joint written out on every assignment of an enumerable instance; the simulator’s label and state marginals are held to enumeration; the gradient identity Equation 40 to central differences; the joint is non-decreasing across every block for every label solver; and label accuracy up to permutation against the planted labels is measured at an equal budget of sweeps for each of the three solvers, so whether a cluster move escapes what iterated conditional modes freezes into is a measurement here and not a premise—and on the planted lattice it is the start, not the move, that decides it.

6.5 Validation

- *The joint written out* on every assignment of the enumerable instance referees the constructed factor graph’s log-density, at equality. It establishes that the three adapters compose into the model Equation 3 states; it establishes nothing about any inference on it.
- *Enumeration* of the same instance referees the simulator’s label and state marginals and the per-class posteriors. It establishes exactness at $2^4 \cdot 2^{12}$ states and stops there.
- *Central differences* referee the gradient identity Equation 40. It establishes the M step’s derivative, not the E step’s posterior, which the enumeration above covers separately.
- *Monotonicity of the joint across every block*, for each of the three label solvers, referees the ascent. It establishes that no block decreases the likelihood; it does not order the solvers, and cannot, since a monotone ascent to a worse fixed point is still monotone.
- *The planted labels*, recovered up to permutation at an equal budget of sweeps, are the referee for the label solvers past enumeration. They establish which solver reaches the planted labelling on this fixture — where the measurement says it is the start and not the move that decides it — and they do not establish that the planted labelling is the maximum of Equation 42, which is what the enumerated MAP labelling on the canonical instance would add.

7 Problem Statement: Large Parsimony

7.1 The model

Parsimony is the phylogenetic criterion with no model: a topology is scored by the fewest state changes that explain the alignment, Equation 20 under unit costs and Equation 21 under a step matrix, with no branch lengths and no probabilities, so the score is an integer where the likelihood is a real number. *Small* parsimony scores a given tree; *large* parsimony is the search,

$$\mathcal{T}^* = \arg \min_{\mathcal{T}} F(\mathcal{T}), \quad F(\mathcal{T}) = \sum_{s=1}^L F_s(\mathcal{T}), \quad (43)$$

over the $(2n - 5)!!$ unrooted topologies [27]. It is NP-hard in general, which is why it is stated as the discrete half of Section 3 with the continuous half removed, and why the same neighbourhoods and the same budget accounting serve it.

Supported sizes. The same sizes as Section 3, with the ceiling set by the same $(2n - 5)!!$ enumeration — five and six taxa per pull request, seven at the stress tier, eight at the release tier, where 10,395 topologies each cost one post-order pass rather than one optimization. A candidate being that much cheaper is the point: at an equal budget of candidates scored, the parsimony climb reaches a neighbourhood an order of magnitude larger than the likelihood climb, and the trade is that the criterion is the wrong one in the region Figure 8 names. The brute force over every labelling of the internal nodes is k^{n-2} per site and obeys the same 200,000-state refusal.

7.2 As a factor graph

One site is the tree factor graph of Section 3 in the min-plus semiring (Figure 7): the pairwise factor on a branch is the cost of a change, 0 on agreement and W_{ij} otherwise, a leaf’s indicator is 0 at its observed state and ∞ elsewhere, and the root prior is constant. Max-product with the sum replaced by a minimum and the product by a sum is Sankoff’s recursion, and under the unit matrix Fitch’s set recursion carries the same minimum. The criterion is therefore the short-branch limit of Equation 18, and Equation 11 is where that limit re-enters the likelihood problem, as an upper bound on $\ell^*(\mathcal{T})$ a search can rank a neighbourhood by.

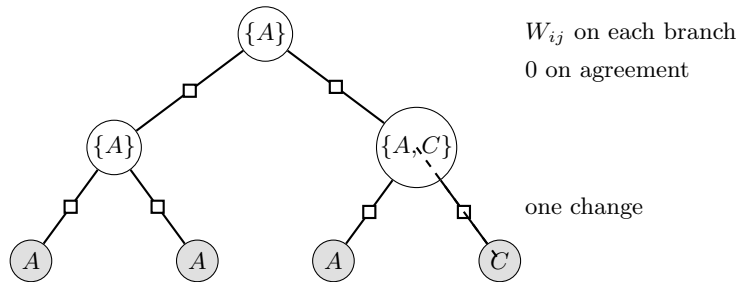


Figure 7: One site of large parsimony as the same tree factor graph in the min-plus semiring. Shaded circles are the observed characters, open circles the Fitch sets of Equation 20 carried up the tree, and squares the branch factors, whose table is 0 on agreement and W_{ij} otherwise (Equation 21). The right-hand cherry disagrees, so its parent takes the union and the site’s score is one change, drawn dashed on the branch that pays it. The structure is Figure 1 with the probabilities replaced by costs and the branch lengths removed; the search over the $(2n - 5)!!$ topologies is what makes the problem large parsimony. A sketch of the structure and not a rendering of any instance.

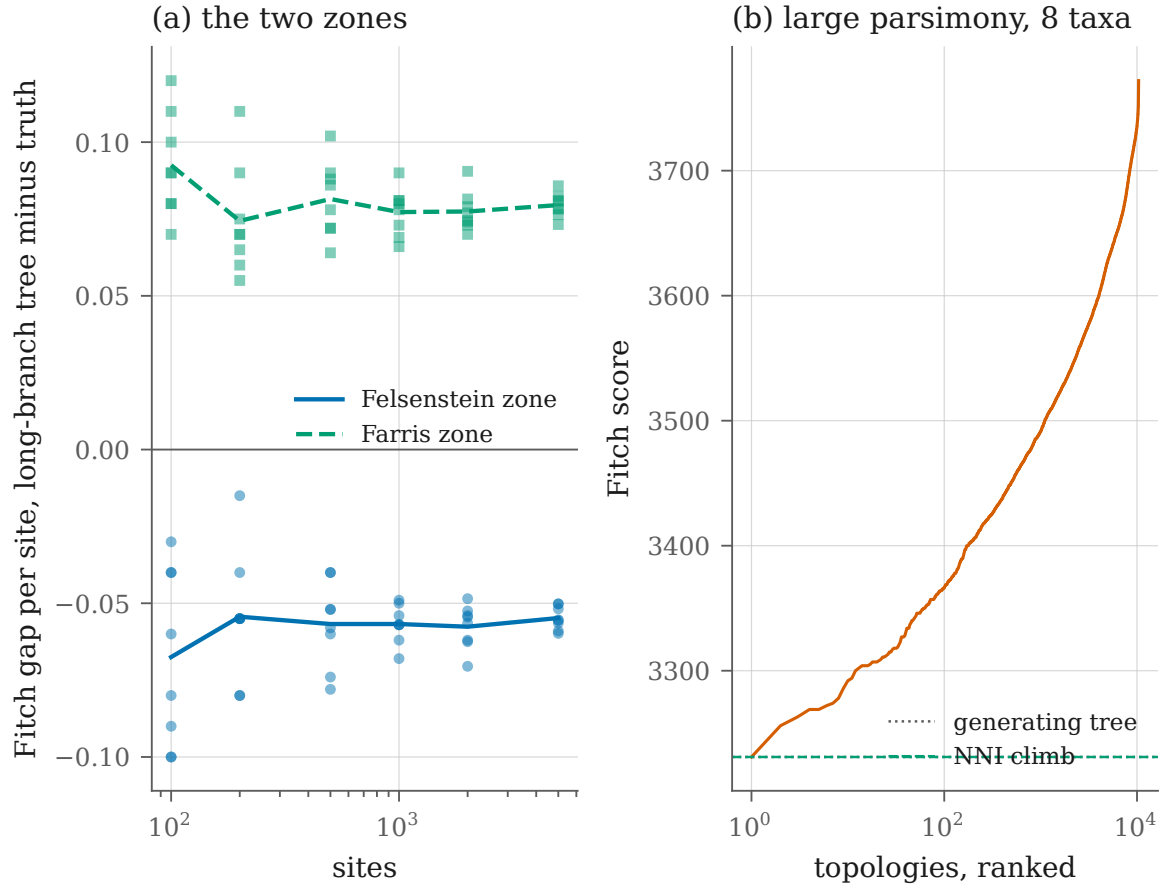


Figure 8: Parsimony in the two zones, and large parsimony on the eight-taxon fixture. (a) Four taxa, pendant branches 0.75 and 0.02 expected substitutions per site with internal branches 0.02, 4-state Jukes-Cantor, seed 20260904, 8 independent alignments at each of 6 site counts from 100 to 5000. Each marker is the Fitch score of the tree grouping the two long branches minus the generating tree's, per site; the line is the mean. In the Felsenstein zone the long branches are not a cherry and the mean gap is -0.068 at 100 sites and -0.055 at 5000: negative, so the criterion prefers the wrong tree, and not closing, which is statistical inconsistency. In the Farris zone the same branches are a cherry and the gap is 0.092 and 0.080. (b) Every one of the 10_395 unrooted topologies on the 8-taxon fixture, scored at 2000 sites (seed 20260904) and ranked; the generating tree scores 3231 and is the minimum. A nearest-neighbour-interchange climb from a random start reaches 3231 after scoring 26 candidates of a 400 budget, the generating tree; the enumeration is the oracle that says so.

7.3 The algorithm: hill climbing on the score

The climb of Section 3, Algorithm 13, with the fit replaced by one post-order pass: a random start by stepwise insertion, the best neighbour under nearest-neighbour interchange or subtree prune-and-regraft taken while one improves, each topology scored at most once, and the budget counted in candidates scored. A candidate costs one pass rather than one optimization, so the same budget reaches a neighbourhood an order of magnitude larger, and the trade is that the criterion it climbs is the wrong one in a region that can be named.

Regime. The step matrix must be a metric, since an unrooted topology has one score only when every rooting scores the same; an asymmetric matrix is refused rather than scored at an arbitrary root. And the criterion is *statistically inconsistent* in the Felsenstein zone (Figure 8): with two long branches that are not a cherry, convergent change on them is cheaper to explain by grouping them than by the true tree, so the error rate goes to one as the alignment grows [27]. Moving the same branches to be a cherry gives the Farris zone, where the criterion is consistent and fast; the pair is what makes a measurement interpretable, because an implementation that is merely broken fails both.

What pins it. The score against a brute force over every labelling of the internal nodes, which shares no traversal with either recursion, at exact equality; Sankoff under the unit matrix against Fitch on every topology of a fixture; and the search against the enumeration of Equation 43's domain from every start where it reaches, with the Felsenstein-zone fixture asserted to return the wrong tree while the likelihood optimum on the same alignment is the right one.

7.4 Validation

- *A brute force over every labelling of the internal nodes*, which shares no traversal with either recursion, referees the score at exact equality. It establishes both recursions on a given tree; it says nothing about the search.
- *Fitch against Sankoff under the unit step matrix*, on every topology of a fixture, referees the two implementations against each other. It establishes that the set recursion and the cost recursion are the same criterion; being two of ours, it is agreement and not truth, which is what the brute force above supplies.
- *Enumeration of Equation 43's domain* referees the climb from every start where it reaches. It establishes that the search found the minimum-score topology at those taxon counts.
- *The Felsenstein-zone fixture* referees the criterion rather than the code: its enumerated parsimony optimum is asserted to be the wrong tree while the likelihood optimum on the same alignment is the right one. It establishes that the implementation reproduces a known statistical inconsistency, which is the one case an implementation that is merely broken cannot pass; it establishes nothing about accuracy outside that zone, for which the Farris-zone pair is kept.
- *The metric condition on the step matrix* is checked as a refusal, not as a result: an asymmetric matrix scores each rooting of an unrooted topology differently and is refused rather than scored at an arbitrary root.

8 Problem Statement: Frustrated Lattices

8.1 The model

Two instances of Equation 25 whose ground-state energy is known from outside, admitted because more than one algorithm is scored against them. The first is the two-state antiferromagnet, $J < 0$, on the periodic triangular lattice: N sites, $3N$ bonds, $2N$ triangles, every bond in exactly two triangles. A triangle is not two-colourable, so each keeps at least one agreeing bond, and counting (triangle, agreeing bond) pairs gives $2A \geq 2N$ for the agreeing bonds A of any configuration; with a zero field and $J = -|J|$,

$$E_{\min} = |J| N, \quad (44)$$

attained, and attained by many configurations: the residual entropy per site tends to Wannier's 0.3231 in the thermodynamic limit, a value reported for orientation and never asserted at a finite size. The second is a planted Viana–Bray spin glass: couplings $\pm J$ on a sparse random graph, built around a state σ^* by setting each coupling to reward what the state does and then inverting a fraction f of them, so that with u inverted bonds

$$E(\sigma^*) = -J(|E| - 2u), \quad E_{\min} \leq E(\sigma^*), \quad (45)$$

a known energy at any size and an upper bound on the ground state, never the ground state itself: at $f = 0$ the instance is a gauge transform of the ferromagnet and trivial, and as f grows the planted state stops being near-optimal (Figure 10).

Supported sizes. Both instances state their answer at every size, so unlike every other section here the size is not bounded by an oracle. The periodic triangular lattice is carried at 3×3 , 4×4 and 9×9 ; the first two are inside the $2^{|V|}$ enumeration, which attains Equation 44 and measures its degeneracy, and the third is past it, where the same closed form still holds. The planted spin glass is carried from ten to a thousand nodes at mean connectivity three and four, with the inverted fraction f swept: at $f = 0$ the instance is a gauge transform of the ferromagnet and measures nothing, and the useful range is where the planted state stops being near-optimal.

8.2 As a factor graph

The factor graph of Section 4 with $\psi_{ij} = e^{J_{ij}\delta(\sigma_i, \sigma_j)}$ and $J_{ij} < 0$ on some or every bond. Nothing changes in the graph; what changes is that the odd cycles of Section 4 are now present, so the ground state is not a colouring and the bond probability $1 - e^{-\beta J_{ij}}$ of the cluster moves is not a probability. Figure 9 is the counting argument drawn on one triangle.

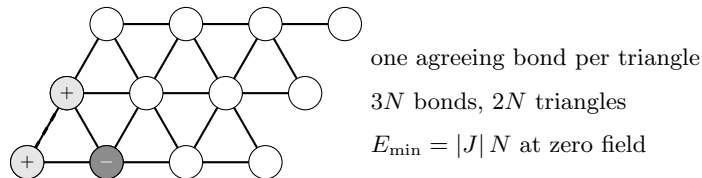


Figure 9: A patch of the periodic triangular antiferromagnet. Every bond wants its endpoints to disagree, and a triangle is not two-colourable, so at least one bond per triangle agrees — the dashed bond of the labelled triangle, where $+$ and $-$ are the two states. Counting (triangle, agreeing bond) pairs over the $2N$ triangles and $3N$ bonds gives Equation 44, a ground-state energy known at every size and attained by many configurations. The planted spin glass of Equation 45 is the same picture on a sparse random graph with a fraction of the couplings inverted. A sketch of the structure and not a rendering of any instance.

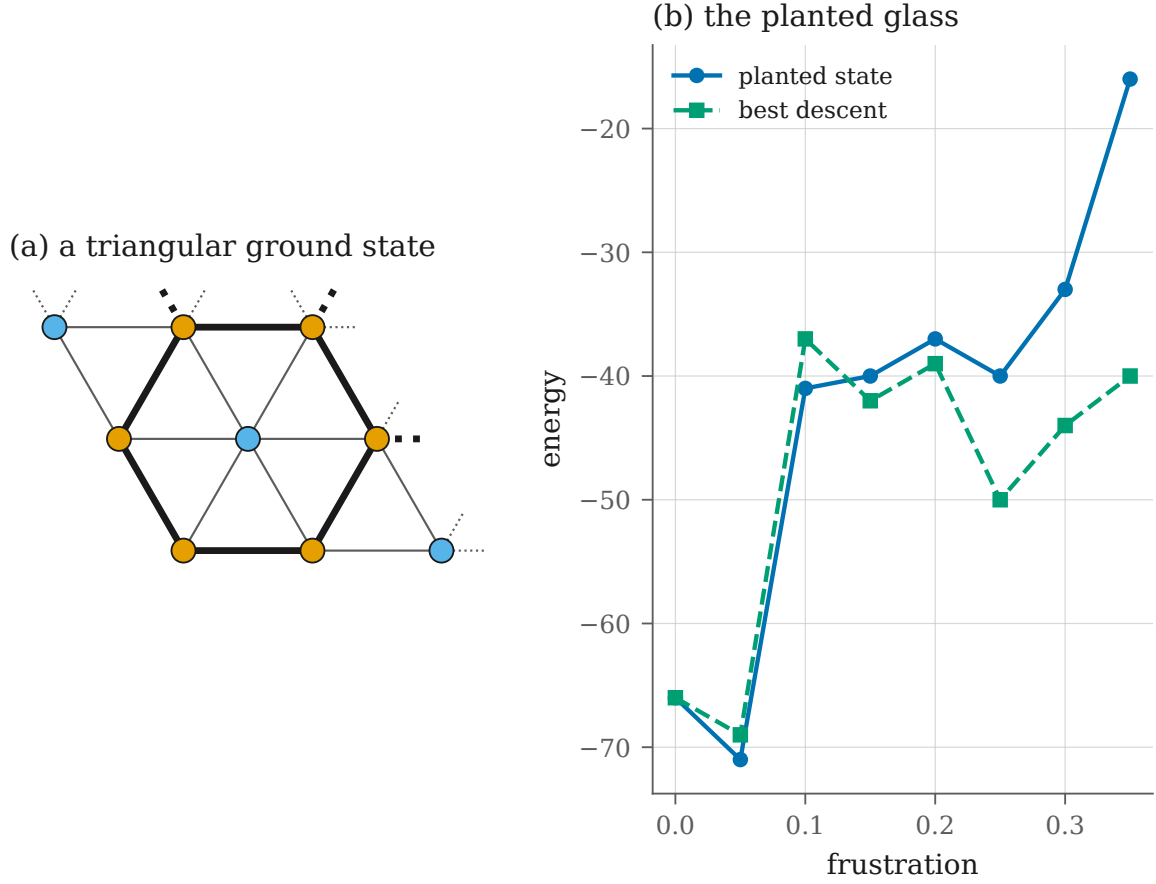


Figure 10: Two frustrated instances with an answer from outside. (a) The 3x3 periodic triangular Ising antiferromagnet, coupling -1, 27 bonds (dotted: bonds wrapping the torus), with one of its enumerated ground states coloured by spin. Heavy bonds join agreeing spins: 9 of 27, one in three, which the double count over triangles fixes as the minimum at every size, so the ground-state energy is 9 exactly. (b) A planted Viana-Bray spin glass on 60 sites at mean degree 4, couplings of magnitude 1, one instance per frustration from seed 20260908: the energy of the planted state against the lowest energy reached by 20 single-site descents from random starts. Descent lands on the planted energy at 3 of 8 frustrations and goes below it at 5, where the planted state is an upper bound on the ground state and not the ground state.

8.3 The algorithms that apply

Enumeration over 2^N configurations and the maximum cut of Section 4.5, where they reach; the semidefinite relaxation of Equation 29, which certifies a ratio at any size; single-site descent, which is Algorithm 8 at zero temperature, and the annealing and parallel tempering of Section 4.8, Algorithm 10, which are the searches these instances exist to separate. The exact cut refuses a negative coupling, since the energy is not submodular, and the cluster moves refuse it for the reason above, so the instances draw the boundary between what is solved exactly and what is only bounded or sampled.

8.4 The initializer

A uniformly random configuration, and restarts of it: on a landscape whose optimum is degenerate, which start is drawn decides which ground state is reached and the count of restarts is part of the budget the comparison charges.

What pins it. Equation 44 at every size, and enumeration attaining it at the sizes enumeration reaches, with the degeneracy asserted and the entropy reported; Equation 45 as an inequality on every search's result, with descent landing on the planted energy at low f and below it at high f measured rather than assumed. Whether annealing or tempering beats restarts is a property of the instance, stated as a measurement at equal sweeps (Section 4.8).

8.5 Validation

- *Equation 44*, a counting argument, referees the triangular ground state at every size. It establishes the minimum energy and that a search attaining it has attained the optimum; it does not identify *which* ground state, the optimum being massively degenerate, so the residual entropy is reported for orientation and never asserted at a finite size.
- *Enumeration over 2^N configurations* referees the same energy where it reaches, and the degeneracy with it. It establishes that the closed form and the code agree at the sizes both cover.
- *Equation 45* referees every search on the glass as an *inequality*: the planted energy is an upper bound on the ground state, never the ground state. It establishes that a search returning a higher energy has failed; it cannot establish that one reaching the planted energy has succeeded, and at a large inverted fraction a search is expected to go below it.
- *The refusals* are checked as results: the exact cut refuses a negative coupling, the energy not being submodular, and the cluster moves refuse it because $1 - e^{-\beta J}$ is not a probability there. They establish that the boundary between exact and approximate is drawn in the code and not only in the text.
- *An equal budget of sweeps* is what compares descent, annealing and tempering here. It establishes which wins *on this instance*, at that budget, over shared seeds; it establishes nothing general about the algorithms, which is why the comparison is reported as an experiment.

9 Problem Statement: The Gaussian Mixture

9.1 The model

Independent observations $y_1, \dots, y_N$, each drawn from one of K components chosen by a weight vector,

$$p(y_i) = \sum_{k=1}^K w_k \mathcal{N}(y_i; \mu_k, \sigma_k^2), \quad \sum_k w_k = 1, \quad w_k > 0, \quad (46)$$

with the component label z_i latent and retained by the simulator so that a clustering has something to be checked against. It is the hidden Markov model of Section 5 with the chain removed: the Gaussian family is the same emission family, and the likelihood inherits its lack of a maximum (Section 13) unchanged.

Supported sizes. One dimension, K components in single figures, N observations from hundreds to hundreds of thousands. The simulator is checked at two components over 200,000 draws, where a component frequency's standard error is under 0.0011; the budgeted comparison of methods runs five components at 1.5 standard deviations of separation over 500 observations at the release tier. The exact clustering cost that referees the seeding is a dynamic programme over cut positions in the sorted observations, which is why the model is carried in one dimension: in higher dimension the same problem is NP-hard and Equation 49 would have nothing to be checked against. Enumerating the K^N labellings is out of reach at every size here, so the responsibilities are refereed by the hidden Markov model's own recursion instead.

9.2 As a factor graph

N variables $x_i \equiv z_i$ with one unary factor each, $\psi_i(z_i) = w_{z_i} \mathcal{N}(y_i; \mu_{z_i}, \sigma_{z_i}^2)$, and no pairwise factor at all. Equation 2 is then a normalization per variable, and the belief at i is the responsibility

$$r_{ik} = \frac{w_k \mathcal{N}(y_i; \mu_k, \sigma_k^2)}{\sum_{k'} w_{k'} \mathcal{N}(y_i; \mu_{k'}, \sigma_{k'}^2)}, \quad (47)$$

the object an HMM's forward-backward pass produces with a recursion and a mixture produces with a division. That the M step receives the same object either way is what lets one emission family serve both, and Figure 11 is Figure 5 with the transition factors removed.

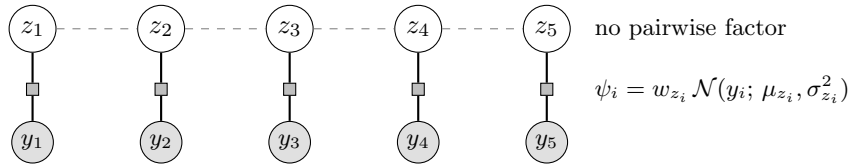


Figure 11: The Gaussian mixture as a factor graph: N component labels z_i over K values, one shaded unary factor each folding in one observation y_i , and no pairwise factor at all. The dashed lines mark where the transition factors of Figure 5 would sit; removing them is exactly what turns the hidden Markov model into the mixture, and it turns Equation 2 into the per-variable normalization that is Equation 47. The same emission family serves both, and receives the same responsibilities from either. A sketch of the structure and not a rendering of any instance.

9.3 The algorithms: expectation-maximization and the gradient fit

Expectation-maximization, Algorithm 14 with the chain removed, alternates Equation 47 with the weighted re-estimate of each component's mean and variance and of the weights as the

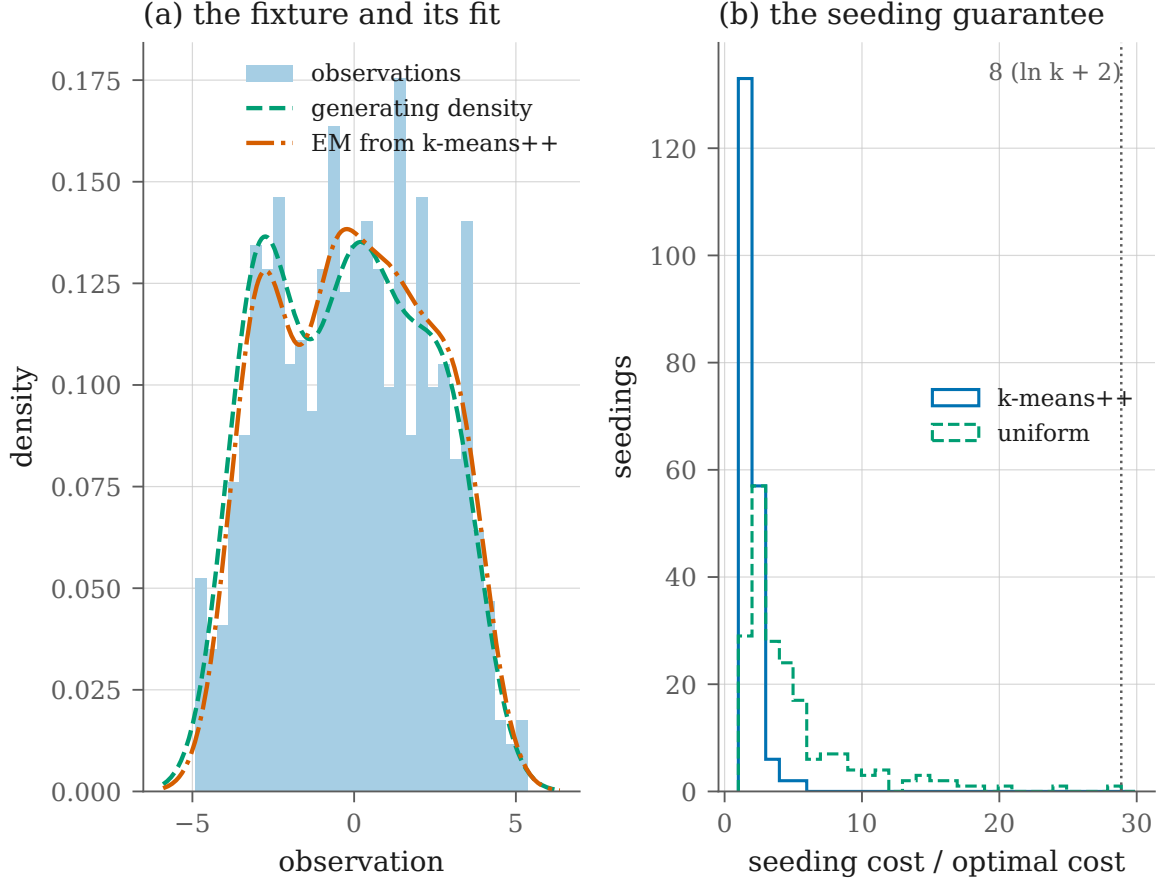


Figure 12: A Gaussian mixture of 5 components with means -3, -1.5, 0, 1.5, 3, unit scales and weights 0.3, 0.1, 0.25, 0.15, 0.2: 500 observations at seed 20260908. (a) The observations, the generating density, and the density expectation-maximization reaches from one k-means++ start at the 500-iteration cap, still moving, at log-likelihood -1122.3. (b) The k-means cost of a seeding divided by the exact optimal cost from the one-dimensional dynamic programme, over 200 independent seedings of the same observations each way: k-means++ averages 1.90 (worst 5.61) and uniform seeding 4.95 (worst 28.68), against the published bound of 28.88 on the k-means++ expectation. The bound is a statement about the mean, and the mean is what is held to it.

mean responsibility, and increases the likelihood at every iterate [55]; the same objective is also fitted by the quasi-Newton optimizer of Section 2.2 over an unconstrained vector, the weights through a softmax and the scales through a logarithm. The two share the model and nothing else, which is what makes their agreement evidence. Where the components overlap the fixed point is approached linearly, so a raw iterate stopped at a count sits above its basin’s optimum by an amount a gradient polish removes, and an equal-budget comparison charges the polish to every method.

9.4 The initializer: k-means++

The first initializer here that reads the data, Algorithm 5. The first centre is an observation drawn uniformly; each subsequent one is an observation y drawn with probability proportional to its squared distance from the nearest centre already chosen,

$$\Pr(c_{j+1} = y) = \frac{D(y)^2}{\sum_{y'} D(y')^2}, \quad D(y) = \min_{j' \leq j} |y - c_{j'}|, \quad (48)$$

and the expected k -means cost of the seeding, before any refinement, is within a factor of the optimum [7]:

$$\mathbb{E}[\phi(c)] \leq 8(\ln K + 2) \phi^*. \quad (49)$$

In one dimension ϕ^* is computable exactly—an optimal clustering partitions the sorted observations into contiguous runs, and a dynamic programme over the cut positions finds it—so the bound has something to be checked against, which it does not in higher dimension where the problem is NP-hard. Seeding is a statement about the cost and not about the likelihood: whether a seeded start reaches a better optimum of Equation 46 than a uniform one is measured over seeds and reported whichever way it falls.

What pins it. The component M step against the hidden Markov model’s on the same responsibilities, at equality; expectation–maximization against the gradient fit at the same optimum, and its iterates monotone; the seeding cost against the exact dynamic programme, with the mean over seedings held to Equation 49; and recovery of the generating parameters up to the permutation of components, with the alignment part of the test. A component whose variance reaches the floor of Section 13 is a refusal, in either algorithm.

9.5 Validation

- *The hidden Markov model’s M step on the same responsibilities* referees the mixture’s, at equality. It establishes that one emission family serves both models; being two of ours it is agreement, not truth.
- *The gradient fit of the same likelihood* referees expectation–maximization at its fixed point, and monotonicity referees every iterate. The two share the model and nothing else, which is what makes their agreement evidence; neither establishes that the shared optimum is global, and where the components overlap the fixed point is approached linearly, so an iterate stopped at a count sits above its basin’s optimum by an amount a polish removes.
- *The exact clustering cost*, a dynamic programme over the cut positions of the sorted observations, referees the seeding of Equation 48 and is the one oracle this section’s initializer has. It establishes that the mean cost over seedings satisfies Equation 49; it says nothing about the likelihood, and whether a seeded start reaches a better optimum than a uniform one is measured over seeds and reported whichever way it falls.
- *Recovery up to the permutation of components* is the referee for the fit itself, with the alignment part of the test. It establishes calibration on data from the model.

- *The variance floor* is checked as a refusal in both algorithms: a component collapsing onto one observation sends the likelihood to infinity, so the fit refuses rather than reports it (Section 13).

10 Problem Statement: Continuous Test Functions

10.1 The functions

Three standard functions with minimizers known in closed form, so the optimizer of Section 2.2 is checked against something that is not a likelihood surface: on a likelihood, a fit that stops early and a parameter the data identify weakly look the same, and these separate the two [66].

$$f_R(x) = \sum_{i=1}^{d-1} \left[b(x_{i+1} - x_i^2)^2 + (a - x_i)^2 \right], \quad x^* = (a, \dots, a), \quad a = 1, \quad b = 100, \quad (50)$$

$$f_{Ra}(x) = 10d + \sum_{i=1}^d \left[x_i^2 - 10 \cos(2\pi x_i) \right], \quad x^* = 0, \quad (51)$$

$$f_H(x, y) = (x^2 + y - 11)^2 + (x + y^2 - 7)^2, \quad (52)$$

the last with four equal minima of value 0, at $(3, 2)$ and three published points. Rosenbrock is a curved valley whose difficulty is conditioning alone; Rastrigin is a quadratic bowl under a cosine ripple with about 10^d local minima, every one satisfying the first-order condition; Himmelblau has no single answer, so “the” optimum is not a well-formed question and a method that always returns the same minimum is reading its start. They are not likelihoods: the Hessian at a minimum is not an observed information, and no interval is built from it. Figure 13 sketches the shape of each.

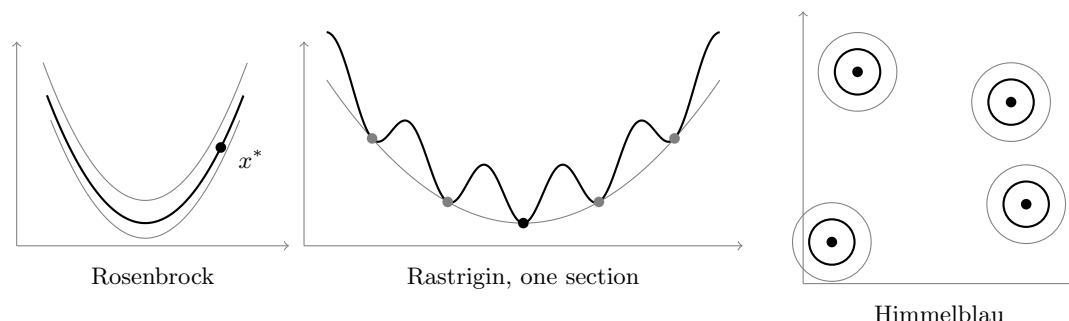


Figure 13: The three surfaces, schematically. Rosenbrock is a single curved valley whose difficulty is conditioning alone, and its minimizer is the marked point of Equation 50. Rastrigin, drawn as one one-dimensional section, is a quadratic bowl under a cosine ripple: the black point is the global minimum of Equation 51 and every grey point a local minimum satisfying the same first-order condition, of which there are about 10^d in d dimensions. Himmelblau has four minima of equal value, Equation 52, so “the” optimum is not a well-formed question and a method that always returns one basin is reading its start. A sketch of the shape of each surface, not a rendering of any of them; the rendered surfaces are Figure 14.

Supported sizes. The size here is the dimension d , and it is the only problem in this document whose oracle costs nothing at any size, the minimizers being closed forms. Rosenbrock is fitted at $d = 2, 3, 4$ and 5 ; Rastrigin at $d = 2$ and 3 , where the roughly 10^d local minima already make a single fit’s success a rate rather than an outcome; Himmelblau is two-dimensional by definition, its four minima published points. There is no per-tier ceiling and no enumeration: what grows

with d is not the cost of the referee but the number of basins a multi-start has to find, which is why the fraction of starts reaching the global minimum is the quantity asserted.

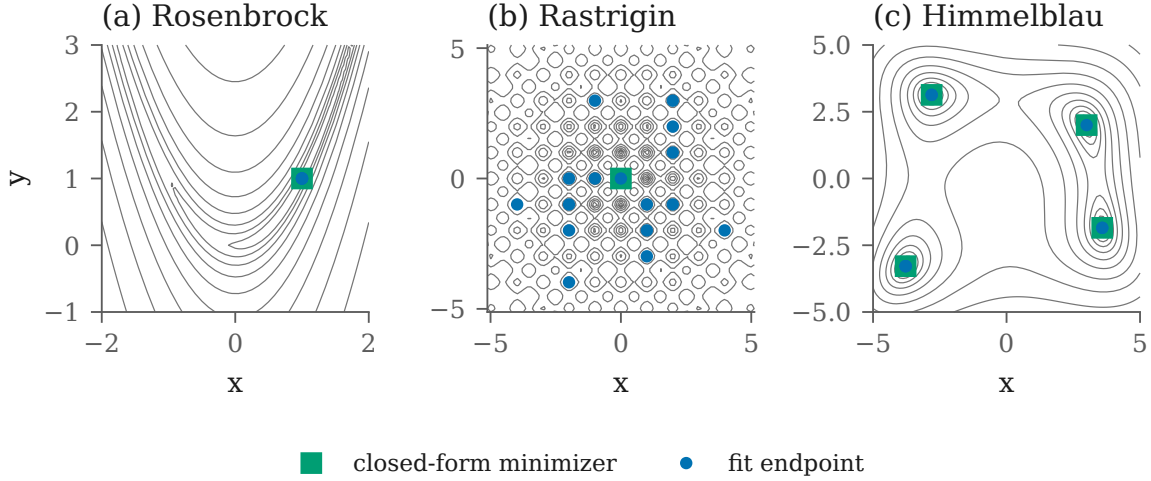


Figure 14: The three continuous test functions in two dimensions, with the minimizers known in closed form and the endpoints of L-BFGS fits from Gaussian random restarts around each function’s own start, the origin for Rastrigin (seed 20260908). (a) Rosenbrock, 4 restarts at scale 1: 4 of 4 reach the minimizer (1, 1). (b) Rastrigin, 16 restarts at scale 2: 1 of 16 reach the global minimum at the origin, every fit stopping at the local minimum of the cell it started in, so a single fit of this surface reports convergence and not optimality. (c) Himmelblau, 8 restarts at scale 3: 8 of 8 reach one of the four equal minima, which one depending on the basin the start fell in. Contours are drawn in $\log(1 + \text{value})$.

10.2 The algorithms that apply

Each function is an instance of the objective abstraction with the identity as its constraint map, so everything that runs on a likelihood runs on it: the quasi-Newton fit, the Hamiltonian sampler of Section 2.3 read as an energy, and annealing with Hamiltonian proposals on a schedule (Section 4.8). They are the continuous counterpart of the frustrated lattices: an equal budget of objective evaluations is spent by restarts of the fit and by an annealed chain with a polishing fit, and which wins is a property of the surface rather than of the algorithms.

10.3 The initializers

The three model-free strategies, Algorithm 7: the function’s own start, that start tilted by a fixed alternating amount for a surface whose nominal start is stationary, and Gaussian restarts around it from a passed-in generator, every start fitted and every fit reported with its spread rather than the best alone (Figure 14). A multi-start that reports one answer for a surface with four basins is the failure the abstraction exists to expose.

What pins it. The fit against the analytic minimizer to a stated tolerance, in several dimensions for Equation 50; the automatic gradient against the hand-written closed form, exactly; every one of Himmelblau’s minima reachable from its own basin; and on Equation 51 a *rate*—the fraction of starts from which a converged fit is the global minimum—asserted low rather than asserted

to succeed, which is the measurement that bounds what a single fit of any multimodal surface here may claim.

10.4 Validation

- *The analytic minimizer* referees a fit directly, to a stated tolerance and in several dimensions for Equation 50. It establishes that the optimizer reaches the right point on a surface where “the right point” is known — which a likelihood cannot offer, and which is the whole reason these functions are carried.
- *The hand-written closed-form gradient* referees the automatic one exactly. It establishes the differentiation, separately from the optimization, so a failure of one is not read as a failure of the other.
- *Each of Himmelblau’s four minima* is required to be reachable from its own basin. It establishes that the multi-start reports a spread rather than one answer; a method that always returns the same minimum is reading its start, and that is the failure this function exists to expose.
- *A rate on Equation 51* — the fraction of starts from which a converged fit is the global minimum — is asserted *low* rather than asserted to succeed. It establishes the bound on what a single fit of any multimodal surface here may claim; it establishes nothing about a particular run.
- *An equal budget of objective evaluations* is what compares restarts against an annealed chain with a polishing fit. It establishes which wins on this surface at that budget, and, as with the frustrated lattices, nothing general about either method.

11 The Problems, and What Applies to Each

Tables 1 and 2 are generated from the problem catalogue and the regression suite, not written: a row is a problem, a column an algorithm family or a kind of referee, and a cell says whether the pair exists and what pins it. Tables 3 to 6 are the same reading taken one method family at a time, in the grouping Appendix D uses, and add to each pairing the tier it is validated at and a sentence on when the family wins on that problem and when it does not — the one hand-written part of the three, and the part that names the experiment where a measurement decided it. A pairing the catalogue carries and no test of either significant kind names is marked *untested* rather than omitted, so the table cannot go quiet about a method that exists. Two readings are load-bearing. The first is the standard: where an oracle is affordable at a size, the method is pinned to it; where none is, recovering the simulated truth—the generating parameters or structure of a seeded fixture, to a stated tolerance or at a stated coverage—is sufficient, and the oracle remains desired. The second is the marks that say where that second case holds, per method and per size tier, so the oracle each would need can be named rather than left implicit: a branch-and-bound bound for trees past the enumerable taxon count, a boundary contraction for lattices past the transfer-matrix width, and for the code the density-evolution threshold of Appendix A.7 extended past the erasure channel.

12 Problem Statement: LDPC Decoding

12.1 The model

A binary linear code of length n is the null space of a parity-check matrix $\mathbf{H} \in \{0, 1\}^{m \times n}$ over $\text{GF}(2)$: the codewords are the c with $\mathbf{H}c = 0$, there are 2^k of them with $k = n - \text{rank } \mathbf{H}$, and

	simulation	exact evaluation	message passing	gradient fit	expectation-maximization	initializer	hill climbing	sampling and tempering	descent and expansion	exact ground state	relaxation	policy learning
Phylogenetic tree, Jukes–Cantor	✓	✓	–	✓	–	✓	✓	✓	–	–	–	✓
Phylogenetic tree, general time-reversible	✓	✓	–	✓	–	◦	✓	–	–	–	–	–
Phylogenetic tree, parsimony	–	✓	–	–	–	–	✓	–	–	–	–	–
Potts chain in a field	✓	✓	–	✓	–	–	–	–	–	–	✓	✓
Potts lattice and Markov random field	✓	–	✓	✓	–	–	–	✓	✓	✓	✓	–
Frustrated Potts: triangular antiferromagnet, planted spin glass	✓	–	–	–	–	–	–	–	–	–	–	–
Low-density parity-check code, Gallager (3,6)	✓	–	✓	–	–	–	–	–	–	–	–	–
Hidden Markov model, six emission families	✓	✓	–	✓	◦	–	–	✓ [†]	–	–	✓	✓
Coupled spatio-sequential model	✓	✓	✓	–	✓	✓ [†]	–	✓	–	–	–	–
Gaussian mixture	✓	◦	–	✓ [†]	✓	◦	–	–	–	–	–	–
Continuous test functions	–	✓	–	✓	–	✓ [†]	–	✓	–	–	–	–

Table 1: The algorithms that apply to each problem, read from the problem catalogue, and what referees each: ✓ a test pinned to an oracle, ✓[†] a test pinned to the simulated truth only, ◦ neither kind of test, and – not applicable or not built. A dagger or a circle is a cell where an oracle is wanted and absent.

	enumeration or brute force	closed form or published value	independent algorithm	planted truth	referee, ci tier	referee, stress tier	referee, release tier
Phylogenetic tree, Jukes–Cantor	✓	–	–	–	oracle	–	oracle
Phylogenetic tree, general time-reversible	✓	–	–	–	oracle	–	truth [†]
Phylogenetic tree, parsimony	✓	–	–	–	oracle	–	oracle
Potts chain in a field	✓	–	–	–	oracle	–	truth [†]
Potts lattice and Markov random field	✓	–	✓	–	oracle	–	truth [†]
Frustrated Potts: triangular antiferromagnet, planted spin glass	–	✓	–	✓	oracle	–	–
Low-density parity-check code, Gallager (3,6)	✓	✓	–	–	oracle	–	truth [†]
Hidden Markov model, six emission families	✓	–	✓	–	oracle	–	truth [†]
Coupled spatio-sequential model	✓	–	–	–	oracle	–	–
Gaussian mixture	–	–	✓	–	oracle	–	truth [†]
Continuous test functions	–	✓	–	–	oracle	–	truth [†]

Table 2: The oracle each problem’s evaluators and searches are pinned to, and, per size tier, what referees the tests that pin them: an oracle, the simulated truth only (dagger: an oracle is wanted), or no test of either kind at that tier (–). A planted truth is an upper bound on the optimum rather than the optimum.

Problem	Tier	Referee	When it wins, and when it does not
Phylogenetic tree, Jukes–Cantor	ci	oracle	Ties the objective’s own start at five and six taxa, every start converging in 22 to 27 evaluations (experiment 006); the joining start is the only one available past the enumerable taxon count, where nothing referees it.
Phylogenetic tree, general time-reversible	–	untested	The log-det distance is additive under any Markov process, so it starts a fit the Jukes–Cantor distance cannot; no test of either significant kind names it as a start, which is why the pairing reads untested.
Phylogenetic tree, parsimony	–	–	–
Potts chain in a field	–	–	–
Potts lattice and Markov random field	–	–	–
Frustrated Potts: triangular antiferromagnet, planted spin glass	–	–	–
Low-density parity-check code, Gallager (3,6)	–	–	–
Hidden Markov model, six emission families	–	–	–
Coupled spatio-sequential model	ci	truth [†]	The annealed data-sampled burn-in beats a uniform start at an equal budget on the planted lattice, where it is the start and not the move that decides the labelling; it is refereed by the planted labels alone.
Gaussian mixture	–	untested	The seeding cost is within the published factor of the exact clustering cost, which is the one oracle this initializer has; whether a seeded start reaches a better likelihood optimum than a uniform one is not pinned by any test of either significant kind, so the pairing reads untested.
Continuous test functions	ci	truth [†]	Multi-start is what makes Himmelblau’s four basins reachable at all; on a unimodal surface it spends the budget for nothing, and it is refereed by the analytic minimizers rather than by an oracle of its own.

Table 3: The initializers each problem carries: the size tier at which the pairing is validated, the kind of referee that validates it, and what the pairing is worth. Tier and referee are read from the suite, as in the two tables above — *oracle* a test pinned to an independent exact answer, *truth[†]* a test pinned to the simulated truth only, *untested* a family the problem carries that no test of either kind names, and – a family it carries no method of. The last column is the one hand-written part of these tables, and names the experiment that measured the comparison where one did.

Problem	Tier	Referee	When it wins, and when it does not
Phylogenetic tree, Jukes–Cantor	ci	oracle	Annealing over topologies returns the flat-prior weight over maximized likelihoods, which hill climbing cannot; it pays a full branch-length fit per step, so it loses wherever only the optimum is wanted.
Phylogenetic tree, general time-reversible	–	–	–
Phylogenetic tree, parsimony	–	–	–
Potts chain in a field	–	–	–
Potts lattice and Markov random field	ci	oracle	Cluster moves slow by $1.9\times$ between extent 8 and 24 where single-site sweeps slow by $3.2\times$, a gap of $2.1\times$ and widening (experiment 001); they refuse a negative coupling, where the single-site sweep is all there is.
Frustrated Potts: triangular antiferromagnet, planted spin glass	–	–	No sampler of its own: the lattice’s samplers apply unchanged, and the cluster moves refuse the negative couplings these instances are built from. What the row adds is the ground-state energy every search is scored against.
Low-density parity-check code, Gallager (3,6)	–	–	–
Hidden Markov model, six emission families	ci	truth [†]	The block sweep draws a whole path from its conditional in one pass, where a single-site sweep would move one position at a time against the chain; it is refereed by the simulated truth alone, an oracle being wanted.
Coupled spatio-sequential model	ci	oracle	The factor-graph sweep is one algorithm over the whole coupled graph and is refereed by enumeration on the canonical instance; it is slower per sweep than the specialized Potts kernel it generalizes.
Gaussian mixture	–	–	–
Continuous test functions	ci	oracle	The annealed Hamiltonian chain with a polishing fit is beaten by plain restarts on the mixture at 3,000 evaluations (experiment 004); on Rastrigin it is the only method that crosses the ripple, and which wins is a property of the surface.

Table 4: The samplers each problem carries: the size tier at which the pairing is validated, the kind of referee that validates it, and what the pairing is worth. Tier and referee are read from the suite, as in the two tables above — *oracle* a test pinned to an independent exact answer, *truth[†]* a test pinned to the simulated truth only, *untested* a family the problem carries that no test of either kind names, and – a family it carries no method of. The last column is the one hand-written part of these tables, and names the experiment that measured the comparison where one did.

Problem	Tier	Referee	When it wins, and when it does not
Phylogenetic tree, Jukes–Cantor	ci	oracle	Hill climbing reaches the enumerated optimum from 12 of 12 starts at six taxa; the learned policy ties it over one feature and wins over seven (experiment 005), while random restarts reach 1.000 at the same budget.
Phylogenetic tree, general time-reversible	ci	oracle	The same quasi-Newton fit over a longer parameter vector, refereed by the same brute-force marginalization; it recovers the rate matrix where the alignment identifies it and refuses an interval where it does not.
Phylogenetic tree, parsimony	ci	oracle	One post-order pass per candidate rather than one optimization, so an equal budget reaches a neighbourhood an order of magnitude larger; it climbs the wrong criterion in the Felsenstein zone.
Potts chain in a field	ci	oracle	The transfer matrix makes the chain’s normalizer exact at every length, so the gradient fit is refereed wherever it runs; nothing here separates it from enumeration, which is also exact at these sizes.
Potts lattice and Markov random field	ci	oracle	The minimum cut is exact and polynomial where the couplings are submodular; above two states only alpha expansion applies, with a factor-of-two bound rather than an optimum, and Max-Cut is bounded and never solved.
Frustrated Potts: triangular antiferromagnet, planted spin glass	–	–	No optimizer of its own. The exact cut refuses a non-submodular energy, so these instances are where descent, annealing and tempering are separated at an equal budget of sweeps and nothing is solved exactly.
Low-density parity-check code, Gallager (3,6)	ci	oracle	Sum-product decodes to the bitwise posterior and min-sum to the most probable codeword; both are exact on a cycle-free code and the Bethe approximation on a loopy one, where min-sum is the cheaper and the weaker and the gap is measured.
Hidden Markov model, six emission families	ci	oracle	Expectation–maximization re-estimates in closed form for every family whose M step has one and wins on cost there; the gradient fit is the only route where it does not, and the two must reach the same optimum. Interval coverage is nominal from two standard deviations of separation upward (experiment 002).
Coupled spatio-sequential model	ci	oracle	Block-coordinate ascent does not decrease the joint at any block, for every label solver; iterated conditional modes freezes into disjoint clusters where the cluster move does not, and which wins is measured at an equal budget rather than assumed.
Gaussian mixture	ci	oracle	Restarts reach the best-known optimum from 7 of 40 starts against tempering’s 4 and annealing’s 1 at 3,000 evaluations (experiment 004); where components overlap the fixed point is approached linearly, so an iterate stopped at a count needs a polish.
Continuous test functions	ci	oracle	The quasi-Newton fit reaches the analytic minimizer in every dimension tried on Rosenbrock; on Rastrigin the fraction of starts reaching the global minimum is asserted low, which bounds what one fit of a multimodal surface may claim.

Table 5: The optimizers each problem carries: the size tier at which the pairing is validated, the kind of referee that validates it, and what the pairing is worth. Tier and referee are read from the suite, as in the two tables above — *oracle* a test pinned to an independent exact answer, *truth*[†] a test pinned to the simulated truth only, *untested* a family the problem carries that no test of either kind names, and – a family it carries no method of. The last column is the one hand-written part of these tables, and names the experiment that measured the comparison where one did.

Problem	Tier	Referee	When it wins, and when it does not
Phylogenetic tree, Jukes–Cantor	ci	truth [†]	A surrogate-ranked neighbourhood reaches its optimum at 5 fits against 312, and changes only what is fitted; it wins nothing where the neighbourhood is small enough to fit exhaustively.
Phylogenetic tree, general time-reversible	–	–	–
Phylogenetic tree, parsimony	–	–	–
Potts chain in a field	ci	oracle	The relaxation is exact at every corner to 1e-11 and supplies a gradient the discrete objective has not; it buys nothing at a size enumeration reaches, and the learned policy beats greedy on the same landscape (experiment 003).
Potts lattice and Markov random field	ci	oracle	Belief propagation is exact on a tree and the Bethe approximation otherwise, so it stands in for the normalizer at sizes enumeration cannot reach; on a loopy graph its error is reported and asserted against nothing.
Frustrated Potts: triangular antiferromagnet, planted spin glass	–	–	–
Low-density parity-check code, Gallager (3,6)	–	–	–
Hidden Markov model, six emission families	ci	oracle	The relaxation of the state path is exact at every corner and gives a gradient over paths; it says nothing at a length past which the path enumeration that referees it cannot run.
Coupled spatio-sequential model	–	–	–
Gaussian mixture	–	–	–
Continuous test functions	–	–	–

Table 6: The surrogates each problem carries: the size tier at which the pairing is validated, the kind of referee that validates it, and what the pairing is worth. Tier and referee are read from the suite, as in the two tables above — *oracle* a test pinned to an independent exact answer, *truth[†]* a test pinned to the simulated truth only, *untested* a family the problem carries that no test of either kind names, and – a family it carries no method of. The last column is the one hand-written part of these tables, and names the experiment that measured the comparison where one did.

the rate is k/n . A low-density code makes $\mathbf{H}$ sparse. Gallager’s regular (d_v, d_c) ensemble [31] puts d_v ones in every column and d_c in every row: $\mathbf{H}$ is d_v bands of n/d_c rows, the first band’s row i covering bits id_c to $(i+1)d_c - 1$ and every other band the first under a uniformly random permutation of the columns, so n is a multiple of d_c and $m = nd_v/d_c$. The rows of one band sum to the all-ones vector, so at least $d_v - 1$ rows are dependent and $k \geq n - m + d_v - 1$. The instance this repository carries is $(3, 6)$ at $n = 19,998$, the multiple of six nearest the 20,000 the ticket names: $m = 9,999$ checks, 59,994 nonzeros, design rate $1/2$.

A codeword is sent through a memoryless binary-input channel and y is received. Three channels are supported, each seen only through its log-likelihood ratio,

$$L_i = \log \frac{p(y_i | c_i = 0)}{p(y_i | c_i = 1)} = \begin{cases} (1 - 2y_i) \log \frac{1-p}{p} & \text{binary symmetric, flip probability } p \\ 0 \text{ if erased, else } \pm \infty \text{ with the sign of } 1 - 2y_i & \text{binary erasure, erasure probability } p \\ 2y_i/\sigma^2 & \text{binary-input Gaussian, } y_i = (1 - 2c_i)\sigma \end{cases} \quad (53)$$

positive when the received symbol favours zero. The posterior over the code is then

$$p(c | y) \propto \mathbb{K}[\mathbf{H}c = 0] \exp(-c \cdot L), \quad (54)$$

since $\log p(y | c) = \sum_i \log p(y_i | c_i) - \sum_i c_i L_i$ and the first sum is the same for every word. Decoding asks two different questions of Equation 54: the *bitwise* MAP, each bit at the mode of its own marginal, which minimizes the bit error rate and need not be a codeword; and the *blockwise* MAP, the single most probable codeword, which minimizes the block error rate. They are different answers, as Section 5 says of the chain, and the fixture that separates them is a test.

Supported sizes. Codes of length $n = 24, 48, 96$ and 996 carry the refereed claims, and the instance the roadmap names is $(3, 6)$ at $n = 19,998$ with $m = 9,999$ checks and 59,994 nonzeros. Three ceilings separate what is refereed by what. Enumerating the 2^k codewords is refused past 200,000 words, so the exact decoding referee reaches $k \leq 17$, which at rate $1/2$ is a code of about 34 bits; Gauss–Jordan elimination gives a systematic encoder up to $n = 512$, past which the all-zero codeword is sent and the argument of Section 12 says nothing is lost; and past both, the referee is the density-evolution threshold of Appendix A.7, which exists for the erasure channel alone. The three channels of Equation 53 run at every length.

12.2 As a factor graph

The variables are the bits, $x_i \equiv c_i \in \{0, 1\}$. The factors are a unary $\psi_i(c_i) = e^{-c_i L_i}$ per bit, the channel folded in as an observation is everywhere in this document, and a parity factor $\psi_a(c_{\partial a}) = \mathbb{K}[\bigoplus_{i \in \partial a} c_i = 0]$ per row of $\mathbf{H}$, a hard constraint of degree d_c whose log table is zero on even-parity assignments and $-\infty$ elsewhere. This is the Tanner graph of the code; a bit sits on d_v parity factors and one unary, so in the Forney form every bit passes through an equality node of degree $d_v + 1$ (Figure 15). The graph has cycles—a random draw of the ensemble does not avoid short ones—so Equation 2 is the Bethe approximation on it, and the exact quantities below come from enumerating the 2^k codewords where $2^k \leq 200,000$ (a $(3, 6)$ code at $n = 24$ has $k \geq 14$) and from the tree schedule on a cycle-free code, where message passing is exact.

12.3 The algorithm: sum-product in the log domain

A binary variable’s message is one number, its log-odds, and Equation 2 on a parity factor has a closed form [31, 55, 75]; the decoder is Algorithm 15. From bit i to check a , and from check a to bit i ,

$$m_{i \rightarrow a} = L_i + \sum_{b \in \partial i \setminus a} m_{b \rightarrow i}, \quad m_{a \rightarrow i} = 2 \operatorname{artanh} \prod_{j \in \partial a \setminus i} \tanh \frac{m_{j \rightarrow a}}{2}, \quad \tilde{L}_i = L_i + \sum_{b \in \partial i} m_{b \rightarrow i}, \quad (55)$$

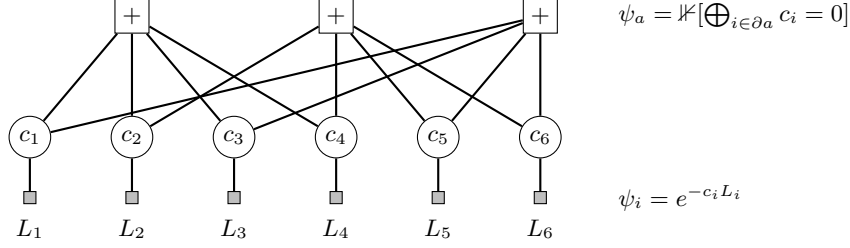


Figure 15: A Tanner graph: bits $c_i \in \{0, 1\}$ as circles, the parity factors of a row of $\mathbf{H}$ as squares, and the channel factor of Equation 53 as a shaded square carrying the log-likelihood ratio L_i at each bit. The drawn graph is six bits and three checks of degree four; the ensemble this repository carries is regular $(3, 6)$, so every bit sits on three parity factors and every parity factor on six bits. The graph has cycles — a random draw of the ensemble does not avoid short ones — so Equation 2 on it is the Bethe approximation and Equation 55 is that approximation in the log domain. A sketch of the structure and not a rendering of any instance.

where the check update follows from $\tanh(L/2) = p(0) - p(1)$ and the fact that the parity of independent bits has a difference of probabilities equal to the product of theirs. The hard decision $\hat{c}_i = \mathbb{K}[\tilde{L}_i < 0]$ is the bitwise MAP under the Bethe beliefs. Replacing the sum by a maximum in Equation 2 gives max-product, whose log-domain check update is

$$m_{a \rightarrow i} = \left(\prod_{j \in \partial a \setminus i} \text{sgn } m_{j \rightarrow a} \right) \min_{j \in \partial a \setminus i} |m_{j \rightarrow a}|, \quad (56)$$

the min-sum decoder, whose decision is the blockwise MAP and is exact on a cycle-free code with a unique maximum; on a loopy code it is the cheaper and the weaker of the two, and the gap is measured rather than assumed.

Three details are load-bearing in the implementation. The schedule is flooding: every check reads the bit messages of the previous half-iteration and every bit reads the check messages just written, one Jacobi sweep per direction over all edges at once, which is the schedule the general implementation runs and so the one it is compared against. The loop stops at the first iteration whose hard decision has every bit decided and satisfies $\mathbf{H}\hat{\mathbf{c}} = 0$, the syndrome check, and otherwise at an iteration cap, which is a decoding failure and the event a block error rate counts. And messages are clipped at $\pm L_{\max}$ with $L_{\max} = 30$ before the check update, because a certain bit's ratio is infinite on the erasure channel and the leave-one-out sums in Equation 55 are formed by subtraction: $\tanh(L_{\max}/2)$ is below one by 1.9×10^{-13} in double precision, and a belief formed under the cap differs from the exact one by less than $e^{-L_{\max}}$ in probability, which the oracle tests absorb.

12.4 The all-zero codeword

Every simulation past the reach of an encoder sends the zero word, and the argument that this loses nothing is standard [75, Sec. 4.1]. Each channel of Equation 53 is output symmetric: sending $c_i = 1$ instead of 0 negates L_i in distribution, and on the symmetric and erasure channels it negates it on every realization, since a flip or an erasure is applied to the bit whatever its value. Both check updates are odd in each argument and the bit update is linear, so negating the ratios at the bits where $c_i = 1$ negates every message and every posterior at exactly those bits, and the hard decision under c is the hard decision under 0 with c added: the error pattern $c \oplus \hat{c}$ has the same distribution whatever codeword was sent. Bit and block error rates measured on the zero word are therefore the rates on any codeword. Where an encoder exists—Gauss–Jordan elimination over $\text{GF}(2)$ to a systematic form, at $n \leq 512$ —the identity is checked per realization on a non-trivial codeword, so the shortcut is itself pinned.

What pins it. Four oracles, none sharing code with the decoder. On a cycle-free code the tree schedule of the general implementation and the enumeration over all 2^k codewords are exact, and the decoder equals both; min-sum equals the general max-product there and returns the enumerated maximum-likelihood codeword, with the margin over the runner-up pinned so that no tie is being broken. On loopy codes the decoder and the general damped flooding are the same Bethe approximation and reach the same fixed point, which is agreement and not truth. The encoder is pinned by $\mathbf{H}c = 0$ and the offsets layout by the dense product it stands in for. And at size, where no enumeration reaches, the closed form is density evolution (Appendix A.7): on the erasure channel the $(3, 6)$ ensemble has a threshold $\epsilon^* = 0.4294$ below which the infinite-length decoder resolves every erasure and above which it does not, and the 19,998-bit code is held to bracketing it.

12.5 Validation

- *Enumeration of the 2^k codewords* referees both decodings on a code short enough to enumerate: the bitwise posterior and the maximum-likelihood codeword, the latter with the margin over the runner-up pinned so that no tie is being broken. It establishes exactness there and reaches no further than the ceiling above.
- *The tree schedule of the general sum-product* referees the decoder on a cycle-free code, where message passing is exact. It establishes that the log-domain specialization of Equation 55 is the same computation as the general implementation; on a loopy code the two agree because they are the same Bethe approximation, which is agreement and not truth.
- $\mathbf{H}c = 0$ *on every realization* referees the encoder, and the dense matrix product referees the sparse offsets layout it stands in for. Each establishes one construction and nothing about decoding.
- *The channel symmetry argument* is checked rather than assumed: where an encoder exists, the error pattern under a non-trivial codeword is compared per realization with the pattern under the zero word. It establishes that the all-zero shortcut is sound at the lengths an encoder reaches, which is what licenses it past them.
- *The erasure threshold $\epsilon^* = 0.4294$* of Appendix A.7 referees the 19,998-bit code, which is held to bracketing it. It establishes the asymptotic behaviour of the ensemble on one channel; it establishes nothing at finite length on the other two, which is the cell Section 11 marks and the extension past the erasure channel would fill.

13 The Properties That Referee Each Method

An algorithm is accepted here against a property that is true independently of this implementation. The properties are not interchangeable, and which one applies is a statement about the method rather than a matter of convenience. Each *What pins it* paragraph above names one of the five kinds below, and the regression suite marks every check with the kind it is: an oracle, a simulated truth, a mathematical invariant, an edge case, or a structural property of the code. A check that fits none of them is either a missing category or a test with nothing to assert.

13.1 Exact answers, where an oracle exists

Where a quantity can be computed a second way that shares no recursion with the first, equality is asserted to a stated tolerance. Direct marginalization over k^{n-2} internal assignments referees pruning; a sum over k^T paths referees the forward recursion; exhaustive enumeration over $q^{|V|}$ configurations referees a Potts marginal; enumeration of $(2n - 5)!!$ topologies referees a search.

Every one is exponential, and that is what makes it an oracle rather than a second opinion. **Two implementations agreeing prove nothing if both are wrong**, so an accelerated method is pinned against the reference and never against another acceleration.

13.2 Structural invariants, where no oracle reaches

Past the sizes enumeration allows, what remains are properties the model must satisfy whatever the answer is:

- rows of a transition matrix sum to one, and $\mathbf{P}(0) = \mathbf{I}$;
- a reversible model satisfies detailed balance, $\pi_i \mathbf{Q}_{ij} = \pi_j \mathbf{Q}_{ji}$;
- the *pulley principle*: on a reversible model the likelihood is invariant to where the root is placed along a branch, so a rooted computation and its re-rooting must agree exactly;
- a gradient matches central differences, relative to the gradient’s own norm—entrywise fails wherever an entry is zero, and an absolute bound does not transfer across data sizes;
- an expectation-maximization iterate increases the likelihood monotonically.

None of these establishes that a method is right. Each establishes that a specific way of being wrong did not happen, which is a weaker and more defensible claim.

13.3 Recovery, which is the acceptance test for a fit

A likelihood that increases proves the optimizer runs, not that the model is right. The acceptance test is to fit data simulated from known parameters and require the intervals to cover the truth at their nominal rate over independent replicates. Two qualifications are load-bearing. Where a model has an exact symmetry—permuting the hidden states of an HMM or the components of a mixture leaves the likelihood unchanged—recovery is stated *up to that symmetry*, and the alignment is part of the test. And coverage is measured as a function of how identifiable the instance is, because a fixture chosen where everything is well separated is a test of the fixture.

13.4 Where the likelihood has no maximum, or no curvature

Two failures are properties of models rather than defects, and both must be handled before they are discovered.

An *unbounded* likelihood has no maximum to find: put a Gaussian component’s mean on a single observation and let its variance go to zero and the density there diverges. A fit that converges there was stopped by its starting point or by a floor, and which one must be knowable—so the bound is explicit, derived from the data rather than chosen, and reaching it is a refusal rather than a clamp.

A *flat* likelihood has a maximum that runs to the boundary: an overdispersed count model approaches its Poisson limit as the dispersion grows, and past the point where the overdispersion is smaller than the sampling noise on measuring it, the data cannot tell one value from another. The two failures are mirror images, and they announce themselves differently—the first as a divergence, the second as a singular information matrix—which is why each is measured rather than inferred from the other.

13.5 Agreement across implementations is a tolerance

An accelerated implementation is pinned against the reference within a declared *relative* tolerance, never bitwise. Relative, because a log-likelihood is a sum over sites and an absolute bound fixed at one size does not transfer to another. Keyed on the *lowest* precision in the comparison,

because a single-precision device cannot be held to a double-precision bound, and one bound loose enough for single precision would let a broken double-precision backend pass. A discrepancy inside the tolerance is not a defect and is not corrected; one outside it is not a tolerance to loosen. The values themselves live beside the measurements they are derived from, not here.

13.6 Published constants, and asymptotic results

Some claims come from outside entirely: the $(2n - 5)!!$ count, Goemans–Williamson’s 0.87856 for maximum cut, alpha expansion’s factor of two, the $8(\ln k + 2)$ bound on k -means++ seeding. These referee an implementation at any size. **No asymptotic result is testable at the sizes an exact oracle allows**, however: a threshold or a limit holds as the problem grows and says nothing where enumeration can check it. Such a result is reported for orientation and the per-instance property is what is asserted.

14 Relevance to the Roadmap

The problem classes are not applications. They are chosen so that each abstraction is tested against something that is not the model it was extracted from: a likelihood engine justified only by trees would be a phylogenetics library, and an optimizer justified only by an HMM would encode a chain.

The order the milestones take follows the dependency between the properties above. Simulation and its ground truth come first, because every later claim is checked against data whose generating parameters are known. The exact evaluators come next, because they are the oracles the accelerated ones are pinned to. Continuous optimization follows, since recovery is the acceptance test and needs both. Discrete search and learned proposals come last, because their acceptance criterion—reaching the optimum enumeration identifies—only exists once enumeration does.

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A Derivations Cited From the Text

Each of these is standard; each is here because a claim in the main text depends on it, and the main text cites rather than derives, for the reader the preface names.

A.1 Pruning is the marginalization over internal states

Root $\mathcal{T}$ at ρ and write z_u for the state at node u , so the leaves’ states at one site are the data $\mathbf{D}_s$ and the internal states are summed. Along each branch the state evolves by the Markov process of Section 3, and the branches are conditionally independent given the state at their common node, so the joint over every node is $p(z) = \pi_{z_\rho} \prod_{v \neq \rho} \mathbf{P}_{z_{\text{pa}(v)} z_v}(t_v)$, and the site likelihood is its sum over the internal states with the leaves clamped—the summand of Equation 17, with k^{n-2} terms. Define $L_u(i) = p(\mathbf{D}_s^{(u)} | z_u = i)$, the probability of the data at the leaves below u given u ’s state. The subtrees hanging from u ’s children share no variable and, given z_u , no factor, so the sum over their states factorizes into one sum per child:

$$L_u(i) = \prod_{v \in \text{children}(u)} p(\mathbf{D}_s^{(v)} | z_u = i) = \prod_{v \in \text{children}(u)} \sum_{j \in \mathcal{A}} \mathbf{P}_{ij}(t_v) p(\mathbf{D}_s^{(v)} | z_v = j),$$

the second equality conditioning on z_v and using that the data below v depends on z_u only through z_v . The inner probability is $L_v(j)$, which is Equation 18; at a leaf $L_u(i) = \mathbb{K}[z_u = i]$ because the state is observed; and summing the root state against its prior gives Equation 19. Nothing in the argument fixes the order of the children or the depth of the tree, so the recursion is a post-order traversal in which each node’s sum is done once per parent state: $O(nk^2)$ per site against k^{n-2} terms. Each L_u is linear in each L_v , so rescaling L_u by a positive c_u scales the site likelihood by $\prod_u c_u$ and nothing else, which is why the logarithm of the scaling may be accumulated apart from the recursion and the rescaled recursion is a transformation of this one rather than a second algorithm [26, 27].

A.2 Forward–backward is the same marginalization on a chain

With the factors of Section 5 the joint over the hidden path and the observations is $p(\mathbf{Z}, \mathbf{D}) = \pi_{z_1} \mathcal{E}_{z_1}(y_1) \prod_{t \geq 1} \mathbf{A}_{z_{t-1} z_t} \mathcal{E}_{z_t}(y_t)$, and the evidence sums it over k^T paths. Define $\alpha_t(i) = p(y_1, \dots, y_t, z_t = i)$. Conditioning on z_{t-1} , and using that $y_1, \dots, y_{t-1}$ depend on z_t only through z_{t-1} , splits the sum over the first $t-1$ states from the rest: $\alpha_t(i) = [\sum_j \alpha_{t-1}(j) \mathbf{A}_{ji}] \mathcal{E}_i(y_t)$, with $\alpha_1(i) = \pi_i \mathcal{E}_i(y_1)$, and the evidence is $\sum_i \alpha_T(i)$: Equation 35. Define $\beta_t(i) = p(y_{t+1}, \dots, y_T | z_t = i)$, so $\beta_T = 1$; conditioning on z_{t+1} in the same way gives the backward recursion of Equation 36. The two meet at any position: $p(z_t = i, \mathbf{D}) = \alpha_t(i) \beta_t(i)$, because the observations before t and after t are independent given z_t , and dividing by the evidence gives the posterior of Equation 36; the same split one step wider gives the pairwise posterior

$p(z_{t-1} = i, z_t = j \mid \mathbf{D}) = \alpha_{t-1}(i) \mathbf{A}_{ij} \mathcal{E}_j(y_t) \beta_t(j) / p(\mathbf{D})$, the sufficient statistic of the expectation step.

The chain is the caterpillar tree of Section 5 rooted at z_1 : each z_t has one child z_{t+1} carrying the branch factor $\mathbf{A}$ and one observed leaf y_t carrying the emission. Appendix A.1 then gives $L_{z_t}(i) = \mathcal{E}_i(y_t) \sum_j \mathbf{A}_{ij} L_{z_{t+1}}(j)$, so $L_{z_t}(i) = \mathcal{E}_i(y_t) \beta_t(i)$ and Equation 19 is $\sum_i \pi_i \mathcal{E}_i(y_1) \beta_1(i) = \sum_i \alpha_1(i) \beta_1(i) = p(\mathbf{D})$: pruning on the chain is the backward pass, and the forward pass is the same recursion run toward the other end with the prior entering as a leaf factor at z_1 . Both passes in the logarithm are the rescaled recursion of Appendix A.1 with the scale taken at every step. Replacing each sum by a maximum leaves every step of the argument intact, since a maximum distributes over a product of non-negative factors as a sum does, and gives Equation 37 [23, 46].

A.3 Sum-product on a tree, and its loopy stationary point

Both derivations above are one fact about a tree-structured factor graph. Take an edge between factor a and variable i and remove it; because the graph is a tree, the component containing a is a subtree $\mathcal{T}_{a \rightarrow i}$ that meets the rest only at x_i . Define the message from a to i as the partition function of that subtree with x_i clamped, $m_{a \rightarrow i}(x_i) = \sum_{x_{\mathcal{T}_{a \rightarrow i} \setminus x_i}} \prod_{b \in \mathcal{T}_{a \rightarrow i}} \psi_b(x_{\partial b})$, and $m_{i \rightarrow a}$ likewise for the subtree on i 's side. The subtrees hanging from a 's other variables $j \in \partial a \setminus i$ are disjoint, so the sum over their variables factorizes into a product of their partition functions, each a function of its own x_j ; and the subtrees hanging from i 's other factors share only x_i , so their product is $m_{i \rightarrow a}(x_i)$. That is Equation 2, with the messages at a leaf variable or leaf factor being 1 and the factor itself. At any variable the whole graph is the union of the subtrees on its factors, so $Z = \sum_{x_i} \prod_{a \in \partial i} m_{a \rightarrow i}(x_i)$ and the marginal is that product normalized; at a factor, $p(x_{\partial a}) \propto \psi_a(x_{\partial a}) \prod_{i \in \partial a} m_{i \rightarrow a}(x_i)$. A message depends only on messages from further from any chosen root, so one leaf-to-root pass computes every message toward it and one root-to-leaf pass every message away, and every marginal follows: two passes, each factor summed once per direction [71, 51, 46]. Pruning is this with a node's state as the variable, a branch matrix as the factor between parent and child, the prior as a factor on the root and a leaf's observation as an indicator factor: $L_u(i)$ is the product of the factor-to-variable messages from below, which is the variable-to-factor message $m_{u \rightarrow \psi_u}$ that u sends up its own branch. Forward-backward is this on the chain. The argument used only that sum distributes over product, so max-product follows by replacing every sum with a maximum, exact on a tree when the maximum is unique.

On a graph with a cycle no edge separates the graph into two subtrees, so the message has no such definition; iterating Equation 2 regardless defines a map on messages, and a fixed point of that map is a stationary point of the Bethe free energy, which Appendix A.4 derives for the pairwise case of Equation 27 and which reads, for factors of any degree,

$$F_{\text{Bethe}} = \sum_a \sum_{x_{\partial a}} b_a (\log b_a - \log \psi_a) - \sum_i (d_i - 1) \sum_{x_i} b_i \log b_i, \quad (57)$$

with d_i the number of factors on i . On a tree $p(x) = \prod_a b_a / \prod_i b_i^{d_i-1}$ exactly, the beliefs being the true marginals, so Equation 57 is $-\log Z$ and the stationary point is the answer of the two passes; on a loopy graph the factorization fails and the fixed point is the approximation [97, 58].

A.4 The Bethe free energy is stationary at a sum-product fixed point

Write the beliefs of a pairwise graph as b_i and b_{ij} and constrain them to be normalized and locally consistent, $\sum_{\sigma_j} b_{ij} = b_i$. Introducing a multiplier $\lambda_{ij}(\sigma_i)$ for each consistency constraint and setting the derivative of Equation 27 to zero gives $b_{ij} \propto \psi_{ij} e^{\lambda_{ij}(\sigma_i) + \lambda_{ji}(\sigma_j)}$ and $b_i \propto \psi_i e^{\sum_{j \in \partial i} \lambda_{ji}(\sigma_i) / (d_i - 1)}$; identifying $e^{\lambda_{ji}(\sigma_i)}$ with $\prod_{l \in \partial i \setminus j} m_{l \rightarrow i}(\sigma_i)$ turns the consistency constraint into Equation 26. So a fixed point of belief propagation is a stationary point of the

Bethe free energy and conversely; on a tree the Bethe free energy is the exact variational free energy, which is why the fixed point there gives $-\log Z$ [97]. The argument uses only that each b_a is a function of the variables its factor touches, so it holds for factors of any degree, with Equation 57 in place of Equation 27 and $d_i - 1$ counting the factors on i beyond the first (Appendix A.3).

A.5 Detailed balance for a cluster move in a field

In the Edwards–Sokal representation the joint over spins and bonds is $p(\sigma, \omega) \propto \prod_{\langle i,j \rangle: \omega_{ij}=1} (1 - e^{-\beta J_{ij}}) \delta(\sigma_i, \sigma_j) \prod_{\omega_{ij}=0} e^{-\beta J_{ij}} \prod_i e^{\beta h_i(\sigma_i)}$, whose spin marginal is the Boltzmann distribution of Equation 25. Drawing bonds given spins and then recolouring a bond cluster $\mathcal{C}$ uniformly are both Gibbs steps at zero field, so the move preserves p there. With a field the recolouring is no longer a Gibbs step: the ratio of the target before and after recolouring $\mathcal{C}$ from μ to ν is $\exp(\beta \sum_{i \in \mathcal{C}} [h_i(\nu) - h_i(\mu)])$, the bond configuration being unchanged, and a uniform proposal accepted with the Metropolis probability of Equation 32 satisfies detailed balance with respect to that ratio. Omitting the accept step gives a chain that runs and mixes and converges to the zero-field distribution, which is why the distributional test is run with a field as well as without.

A.6 The exchange ratio in parallel tempering

The replicas’ joint target is the product $\prod_r \exp(-\beta_r E(\sigma_r))$. Swapping the configurations of replicas i and j changes it by $\exp(-\beta_i E_j - \beta_j E_i) / \exp(-\beta_i E_i - \beta_j E_j) = \exp[(\beta_i - \beta_j)(E_i - E_j)]$, and the proposal is symmetric, so the Metropolis acceptance is Equation 33. Each replica’s marginal under the joint is its own tempered Boltzmann distribution whatever the exchanges do, which is the property the per-replica goodness-of-fit test asserts.

A.7 Density evolution on the erasure channel

On a cycle-free (d_v, d_c) graph the messages arriving at a node are independent, so the erasure probability of a message is a number rather than a distribution and Equation 55 reduces to a scalar recursion. A check-to-bit message is known only if every other incoming bit message is known; a bit-to-check message is erased only if the channel and every other incoming check message are. With x_ℓ the probability that a bit-to-check message is erased after ℓ iterations,

$$x_0 = \epsilon, \quad x_\ell = \epsilon \left(1 - (1 - x_{\ell-1})^{d_c-1}\right)^{d_v-1}, \quad (58)$$

the recursion Richardson and Urbanke [75, Sec. 3.12] derives and names density evolution. The right-hand side is increasing in $x_{\ell-1}$ and in ϵ , so x_ℓ is non-increasing and its limit is the largest fixed point in $[0, \epsilon]$; the threshold ϵ^* is the largest ϵ for which that fixed point is zero, found by bisection, and for $(3, 6)$ it is 0.4294, with 0.3834 for $(4, 8)$ and 0.5176 for $(3, 5)$. A finite code is not cycle-free, but the neighbourhood of a bit to depth ℓ is a tree with probability tending to one as n grows at fixed ℓ , which is why the recursion is the limit of the flooding decoder and why the code at size is held to bracketing ϵ^* rather than to a per-draw agreement—the rule this repository states for every asymptotic result (Section 13).

A.8 The delta method for an interval at a fit

At a maximum $\hat{\theta}$ of the log-likelihood, the observed information $\mathcal{I} = -\nabla^2 \log p(\mathbf{D} \mid \hat{\theta})$ gives the asymptotic covariance $\Sigma = \mathcal{I}^{-1}$ of $\hat{\theta}$. A constrained parameter $\phi(\theta)$ is a smooth function of θ , so to first order $\text{Var}(\phi(\hat{\theta})) \approx J \Sigma J^\top$ with $J = \partial \phi / \partial \theta$ at $\hat{\theta}$. The statement is asymptotic in the sample size and requires $\mathcal{I}$ to be invertible; a singular or ill-conditioned information is a parameter the data does not identify, and the interval is refused rather than reported (Section 13).

B Derivations for the Actor–Critic

Any state-dependent baseline leaves the estimator unbiased. For a function b of the state alone, $\mathbb{E}_{a \sim \pi(\cdot|s)}[\nabla \log \pi(a|s) b(s)] = b(s) \sum_a \nabla \pi(a|s) = b(s) \nabla \sum_a \pi(a|s) = 0$, so subtracting $b(s_t)$ from G_t in (6) changes no expectation. The variance of the per-step term $\nabla \log \pi \cdot (G_t - b)$ is minimized over constant b by a weighted mean of G_t , and over state-dependent b the value $V^\pi(s_t)$ is the choice that removes the part of G_t predictable from the state; a fitted critic approximates it, and its error costs variance, never bias, as long as the fit does not read the action sampled at that step [87].

The generalized advantage telescopes to its two limits. With $\delta_t = R_t + V(s_{t+1}) - V(s_t)$ and $V(s_T) = 0$, $\sum_{l=0}^{T-t-1} \delta_{t+l} = \sum_{u \geq t} R_u - V(s_t) = G_t - V(s_t)$, which is (7) at $\lambda = 1$; at $\lambda = 0$ only δ_t survives. The exponential weights interpolate the two, and the bias of $A_t^{(\lambda)}$ as an estimate of the true advantage is $O(\lambda^k)$ in the critic’s error at k steps ahead [82].

PUCT as policy improvement. With exact leaf values and one decision of depth, every backed-up value in (9) is $Q^*(s, a)$ exactly, and as $N(s) \rightarrow \infty$ the bonus of every action vanishes relative to the gap between the best mean and the rest, so the most visited action is an argmax of Q^* . At depth the means are of returns under the search’s own descents, which concentrate on the children it rates highest; the normalized visit counts are then a distribution whose one-step expected value under Q^π is at least the prior’s, which is the policy-improvement step of policy iteration with the prior as π and the visit distribution as the improved policy [87, 84]. The regression suite checks both on the Potts chain: the argmax exactly at depth one, and the improvement on 13 of 14 sampled states at depth three, the exception being a state whose prior already puts most of its mass on an optimal move.

The clipped objective at the collecting policy. With $\pi = \pi_{\text{old}}$, $\rho_t = 1$ for every decision, so both arguments of the minimum in (8) equal A_t and $\nabla \rho_t = \nabla \log \pi(a_t | s_t)$; the gradient of (8) is $\sum_t \nabla \log \pi(a_t | s_t) A_t$, the actor–critic’s estimator. Away from the collecting policy the minimum removes the gradient wherever the ratio has left $[1 - \epsilon, 1 + \epsilon]$ in the direction the advantage would push it, which is the pessimistic bound on the surrogate improvement that licenses the reuse of a batch [83].

C Derivations of the Bounds

The plug-in bound, (10). $\ell^*(\mathcal{T})$ is a maximum over the feasible set $\tau \geq 0$, and $\hat{\tau}$ lies in it by construction: projected gradient descent on the convex least-squares objective clamps every iterate to the non-negative orthant, so the value at $\hat{\tau}$ is one of the values maximized over. Equality holds exactly when $\hat{\tau}$ is a maximizer.

The parsimony bound, (11). Write $a_e = e^{-kt_e/(k-1)} \in [0, 1]$. Each entry of $\mathbf{P}(t_e)$ is affine in a_e and each branch appears exactly once in the product of (18), so the site likelihood is a polynomial of degree at most one in each a_e : multilinear. A multilinear function on a box attains its maximum at a vertex, where each $\mathbf{P}_e$ is either $\mathbf{I}$ (no change permitted) or $\mathbf{J}/k$ (the child’s state independent of the parent’s, at weight $1/k$). Root the tree at leaf 1, which the pulley principle permits for a reversible model. At a vertex with cut set C (the branches carrying $\mathbf{J}/k$), the site likelihood is a sum over ancestral assignments constant on each component of $\mathcal{T} \setminus C$: $\pi(x_{1,s}) k^{-|C|} k^{\ell'}$, with ℓ' the number of components containing no leaf (each contributes a free sum over k states), provided every leafy component is monochromatic, and zero otherwise. Merging each leafless component into a neighbour across one of its boundary branches removes

one cut and one leafless component at a time and leaves a valid partition with $|C| - \ell'$ cuts, all of whose components are leafy and monochromatic; assigning each its leaf state gives an ancestral labelling with at most $|C| - \ell'$ changes, so $F_s \leq |C| - \ell'$ by minimality of the Fitch score. Hence every vertex value is at most $\pi(x_{1,s}) k^{-F_s}$, and so is the maximum over the box. Summing over sites gives (11); the regression suite enumerates the $2^{|E|}$ vertices at five taxa and finds the bound attained on every site. The entrywise limit $\sup_t \mathbf{P}_{ij}(t)$, one on the diagonal and $1/k$ off it, gives a bound this one dominates, because that matrix is not row-stochastic and the polynomial is increasing in every entry.

The mean-field bound, (12). For any distribution q on configurations, $\log Z = \mathbb{E}_q[-H] + H(q) + \text{KL}(q \| p)$, and the divergence is non-negative (Gibbs' inequality). Restricting q to products and evaluating the expectation gives the right-hand side; the fixed-point iteration $q_i \propto \exp(h_i + \sum_{j \in \partial i} J_{ij} q_j)$ ascends it coordinate-wise, and the bound holds at every iterate, so a truncated iteration is a looser bound and never a wrong one [55, 58].

The spanning-tree bound, (13). $\log Z(\theta)$ is the log-normalizer of an exponential family and therefore convex in θ . For trees T with weights $\rho(T)$ and parameters $\theta(T)$ with $\sum_T \rho(T) \theta(T) = \theta$, Jensen gives $\log Z(\theta) \leq \sum_T \rho(T) \log Z(\theta(T))$. The implementation takes one tree per edge, built by breadth-first search from that edge so it contains it, weighted uniformly; an edge appearing in a fraction ρ_e of the trees carries J_e/ρ_e on each tree containing it and 0 elsewhere, so the weighted mean is J_e , and the fields are shared. Each $\log Z(\theta(T))$ is exact by the leaf-to-root recursion, in the log domain. The bound is tight when the graph is a tree [92].

The ground-state bracket, (14). $Z(\beta) = \sum_{\sigma} e^{-\beta H(\sigma)}$ has k^N terms, each at most $e^{-\beta E_{\min}}$ and at least one equal to it, so $-\beta E_{\min} \leq \log Z(\beta) \leq N \log k - \beta E_{\min}$. Combining with $L \leq \log Z(\beta) \leq U$ and dividing by $\beta > 0$ gives the bracket; as β grows the $N \log k$ term is divided away, until the bounds on $\log Z$ themselves loosen.

D Algorithms Cited From the Text

The algorithms each problem statement cites, grouped by what each is for: initializers, which choose where a search starts; samplers, which draw from a distribution at a temperature; optimizers, which move toward an optimum; and surrogates, which stand in for an expensive evaluation. They are collected here rather than at the point of use because most of them apply to more than one problem, and a statement of an algorithm belongs where it can be cited from each. Four algorithms stay in the body, because there the text derives them as it states them: forward simulation, Algorithm 1; neighbor joining, Algorithm 2; the Wolff move in an external field, Algorithm 3; and the data-sampled burn-in, Algorithm 4.

D.1 Initializers

Algorithm 5 k -means++ seeding of K centres, Equation 48

- 1: draw c_1 uniformly from the observations $y_1, \dots, y_N$
 - 2: **for** $j = 1$ to $K - 1$ **do**
 - 3: $D(y) \leftarrow \min_{j' \leq j} |y - c_{j'}|$ for every observation y
 - 4: draw $c_{j+1} = y$ with probability $D(y)^2 / \sum_{y'} D(y')^2$
 - 5: **end for**
 - 6: **return** the centres $c_1, \dots, c_K$
-

Algorithm 6 The closest tree of a Hadamard spectrum, Equation 24

- 1: recode the alignment to two states and index the 2^{n-1} patterns by the set of taxa differing from a reference taxon
 - 2: $s \leftarrow$ the observed pattern frequencies; $q \leftarrow H^{-1} \log(Hs)$
 - 3: $\mathcal{S} \leftarrow \emptyset$
 - 4: **for** each split in decreasing order of its weight in q **do**
 - 5: **if** the split is compatible with every split in $\mathcal{S}$ **then**
 - 6: $\mathcal{S} \leftarrow \mathcal{S} \cup \{\text{split}\}$
 - 7: **end if**
 - 8: **stop** once $|\mathcal{S}| = n - 3$
 - 9: **end for**
 - 10: **return** the topology whose internal edges are $\mathcal{S}$, with the corresponding weights as branch lengths
-

Algorithm 7 Multi-start from a generator, at a declared budget

- 1: $\theta_0 \leftarrow$ the objective's own start, or that start tilted by a fixed alternating amount where it is stationary
 - 2: **for** $r = 1$ to R **do**
 - 3: $\theta_r \leftarrow \theta_0 + \varepsilon_r$, ε_r drawn from the passed-in generator
 - 4: **end for**
 - 5: fit from every θ_r , each fit charged to the shared budget of objective evaluations
 - 6: **return** every fit with its spread, and the best of them — never the best alone
-

D.2 Samplers

Algorithm 8 One heat-bath sweep at inverse temperature β , Equation 31

- 1: **for** each node i of $\mathcal{G}$, in a fixed order **do**
 - 2: $p(\sigma) \propto \exp(\beta[\sum_{j \in \partial i} J_{ij} \delta(\sigma, \sigma_j) + h_i(\sigma)])$ for $\sigma \in \{1, \dots, q\}$
 - 3: draw $\sigma_i \sim p$ $\triangleright \beta \rightarrow \infty$ is the argmax, which is iterated conditional modes
 - 4: **end for**
 - 5: **return** the configuration after every node has been updated once
-

D.3 Optimizers

D.4 Surrogates

E Reference Taxonomy

Grouped as the repository groups them, so one routing table serves the code and the documents alike. Within a group, books come first, then reviews, then the papers an algorithm implements or a proposal needs as its oracle.

Infrastructure (build, structure, and speed). Software craft: Martin [56], Blandy et al. [15], Ramalho [73]. Systems and hardware, including memory hierarchies and parallel kernels: Bryant and O'Hallaron [19], Hwu et al. [43]. Python performance: Gorelick and Ozsvald [36], Antão [5].

Optimization (discrete and continuous). Algorithms and discrete mathematics: Cormen et al. [21], Rosen [76], Papadimitriou and Steiglitz [69]; the papers behind the cut-based solvers,

Algorithm 9 A cluster move at zero field, Edwards–Sokal bonds

- 1: refuse the move if any $J_{ij} < 0$, where $1 - e^{-\beta J_{ij}}$ is not a probability
 - 2: **for** each edge $\langle i, j \rangle$ with $\sigma_i = \sigma_j$ **do**
 - 3: place a bond with probability $1 - e^{-\beta J_{ij}}$
 - 4: **end for**
 - 5: $\mathcal{C} \leftarrow$ the bond-connected components; in Wolff, the one component containing a uniformly drawn root
 - 6: **for** each component in $\mathcal{C}$ **do**
 - 7: draw a new label ν uniformly from $\{1, \dots, q\}$
 - 8: recolour the component to ν , accepted with Equation 32 where a field is present
 - 9: **end for**
-

Algorithm 10 Parallel tempering over R replicas, Equation 33

- 1: $T_1 < \dots < T_R$; replica r draws from its own generator, spawned from one parent that draws only the exchange uniforms
 - 2: **for** each sweep of the budget **do**
 - 3: **for** $r = 1$ to R **do**
 - 4: advance replica r by one sweep at $\beta_r = 1/T_r$ $\triangleright$ Algorithm 8 or 9
 - 5: **end for**
 - 6: **for** each adjacent pair $(r, r+1)$ **do**
 - 7: exchange the configurations with probability $\min\{1, \exp[(\beta_r - \beta_{r+1})(E_r - E_{r+1})]\}$
 - 8: **end for**
 - 9: record the cold replica's state, and the acceptance of every pair
 - 10: **end for**
 - 11: **return** the lowest energy seen, and the cold replica's recorded weights, Equation 34
-

Algorithm 11 One block sweep over a chain, given the rest of the model

- 1: run the forward recursion Equation 35 over the chain, keeping every α_t
 - 2: draw $z_T \sim \alpha_T$, normalized
 - 3: **for** $t = T - 1$ down to 1 **do**
 - 4: draw z_t with probability $\propto \alpha_t(z_t) \mathbf{A}_{z_t z_{t+1}}$
 - 5: **end for**
 - 6: **return** the path, one exact draw from its conditional and not a sequence of single-site updates
-

Algorithm 12 Alpha expansion, Equation 30

- 1: $\sigma \leftarrow$ any labelling
 - 2: **repeat**
 - 3: **for** each label $\alpha \in \{1, \dots, q\}$ **do**
 - 4: build the two-state graph in which every site keeps σ_i or switches to α
 - 5: $\sigma' \leftarrow$ the labelling of its minimum s - t cut
 - 6: **if** $H(\sigma') < H(\sigma)$ **then** $\sigma \leftarrow \sigma'$
 - 7: **end if**
 - 8: **end for**
 - 9: **until** no expansion lowers the energy
 - 10: **return** σ $\triangleright$ within the constant of Equation 30 of the optimum, provided each expansion was solved exactly
-

Algorithm 13 Hill climbing over a structural neighbourhood

```
1:  $x \leftarrow$  a start;  $\text{score}(x) \leftarrow$  its full evaluation
2: repeat
3:    $\mathcal{N} \leftarrow \mathcal{N}(x)$ , each candidate scored at most once and charged to the budget
4:    $x' \leftarrow \arg \max_{y \in \mathcal{N}} \text{score}(y)$ 
5:   if  $\text{score}(x') > \text{score}(x)$  then  $x \leftarrow x'$ 
6:   end if
7: until no neighbour improves, or the budget is spent
8: return  $x$ , its score always a full evaluation and never a surrogate's
```

Algorithm 14 Expectation–maximization on a chain, or without one

```
1: repeat
2:   E step: the posteriors of Equation 36 by forward–backward; on a model with no chain,
   the responsibilities of Equation 47 by a division
3:   M step: re-estimate  $\pi$  and  $\mathbf{A}$  in closed form from the posterior counts, and the emission
   parameters by the family's own rule — itself an optimization where no closed form exists
4:   assert the likelihood did not decrease
5: until the increase falls below a tolerance, or the iteration cap is reached
```

Algorithm 15 Sum-product decoding in the log domain, Equation 55

```
1:  $m_{b \rightarrow i} \leftarrow 0$  for every edge;  $L_i$  from Equation 53
2: for iteration 1 to the cap do
3:    $m_{i \rightarrow a} \leftarrow \text{clip}(L_i + \sum_{b \in \partial i \setminus a} m_{b \rightarrow i})$  for every edge, at once
4:    $m_{a \rightarrow i} \leftarrow 2 \operatorname{artanh} \prod_{j \in \partial a \setminus i} \tanh(m_{j \rightarrow a}/2)$ , or Equation 56 for min-sum
5:    $\tilde{L}_i \leftarrow L_i + \sum_{b \in \partial i} m_{b \rightarrow i}$ ;  $\hat{c}_i \leftarrow \mathbb{K}[\tilde{L}_i < 0]$ 
6:   stop if every bit is decided and  $\mathbf{H}\hat{\mathbf{c}} = 0$ 
7: end for
8: return  $\hat{\mathbf{c}}$ , and whether the syndrome check was reached — a failure here is the event a block
   error rate counts
```

Algorithm 16 Ranking a neighbourhood by a surrogate, Section 2.7

```
1:  $\mathcal{N} \leftarrow \mathcal{N}(x)$ 
2:  $\hat{s}(y) \leftarrow$  the surrogate's value at every  $y \in \mathcal{N}$ , one cheap pass each
3: if the surrogate is an analytic bound then
4:   discard every  $y$  whose bound cannot beat the incumbent ▷ certified: no violation is
   permitted on any structure an oracle can score
5: else
6:   shift  $\hat{s}$  by a quantile of the held-out residuals, to the coverage the predictor claims
7: end if
8: evaluate in full the best  $\kappa$  candidates by  $\hat{s}$ , and the accepted move in full whatever its rank
9: return the accepted move, with a reported score that is always a full evaluation
```

Kolmogorov and Zabih [47] on which energies a cut minimizes and Boykov and Kolmogorov [17] on the max-flow they run. Numerical optimization: Nocedal and Wright [66], Boyd and Vandenberghe [16] for the semidefinite certificate; Hansen and Ostermeier [38] for the covariance-adapting evolution strategy proposed as a population baseline. Probabilistic inference and message passing: MacKay [55], Koller and Friedman [46], Frey [30], Ortega [67], Bishop [13], Murphy [61]; the review Wainwright and Jordan [91]; the papers Wainwright et al. [92] and Wainwright et al. [93] for the tree-reweighted bounds, and Hsu et al. [42] for spectral initialization of a hidden Markov model. Statistical physics and Monte Carlo: Mézard and Montanari [58], Newman and Barkema [64], Krauth [50], Landau and Binder [52]; the reviews Betancourt [12] and Schollwöck [79]; the paper Machta [54] for population annealing. Information theory and geometry: Amari [3], Cover and Thomas [22]. Learning and reinforcement learning: Goodfellow et al. [35], Prince [72], Sutton and Barto [87], Lapan [53], Raschka [74]; the papers Sutton et al. [88] and Bacon et al. [10] for options, and Schrittwieser et al. [80] for planning with a learned model. Learning for combinatorial optimization: the reviews Bengio et al. [11] and Mazyavkina et al. [57]; the papers Khalil et al. [44], Kool et al. [48], Paulus et al. [70], Henderson et al. [39], Agarwal et al. [1], and the pair Schuetz et al. [81] and Angelini and Ricci-Tersenghi [4].

Application (the science). Phylogenetics: Felsenstein [27], Durbin et al. [23], Compeau and Pevzner [20], Pachter and Sturmfels [68], Yang [96]; the papers Whidden and Matsen [94], Azouri et al. [9], Zhang and Matsen [98], Zhang and Matsen [99], Speyer and Sturmfels [85], Altekari et al. [2], and the external baselines Minh et al. [59] and Kozlov et al. [49]; distance and spectral methods: Saitou and Nei [77], Atteson [8], Steel [86], Hendy and Penny [40], Mossel and Roch [60], with Hsu et al. [42] as the bridge. Information and coding: Blahut [14], Richardson and Urbanke [75], with Nielsen and Chuang [65] as background only; the papers Gallager [31] and Nachmani et al. [62].